

Gamepage\medquest.js

```

1  function Game() {
2      var die1;
3      var die2;
4      var areDiceRolled = false;
5
6      var auctionQueue = [];
7      var highestbidder;
8      var highestbid;
9      var currentbidder = 1;
10     var auctionproperty;
11
12     this.rollDice = function() {
13         die1 = Math.floor(Math.random() * 6) + 1;
14         die2 = Math.floor(Math.random() * 6) + 1;
15         areDiceRolled = true;
16     };
17
18     this.resetDice = function() {
19         areDiceRolled = false;
20     };
21
22     this.next = function() {
23         if (!p.human && p.money < 0) {
24             p.AI.payDebt();
25
26             if (p.money < 0) {
27                 popup("<p>" + p.name + " is bankrupt. All of its assets will be turned
over to " + player[p.creditor].name + ".</p>", game.bankruptcy);
28             } else {
29                 roll();
30             }
31         } else if (areDiceRolled && doublecount === 0) {
32             play();
33         } else {
34             roll();
35         }
36     };
37
38     this.getDie = function(die) {
39         if (die === 1) {
40
41             return die1;
42         } else {
43
44             return die2;
45         }
46     };
47
48
49     function updateBoardUI() {
50         for (let i = 0; i < square.length; i++) {
51             let cell = document.getElementById("cell" + i + "name");
52             if (cell) {
53                 cell.innerHTML = square[i].image ? `` : square[i].name;
54             }
55         }

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56     }
57
58
59     // Auction functions:
60
61
62
63     var finalizeAuction = function() {
64         var p = player[highestbidder];
65         var sq = square[auctionproperty];
66
67         if (highestbid > 0) {
68             p.pay(highestbid, 0);
69             sq.owner = highestbidder;
70             addAlert(p.name + " bought " + sq.name + " for ATP" + highestbid + ".");
71         }
72
73         for (var i = 1; i <= pcount; i++) {
74             player[i].bidding = true;
75         }
76
77         $("#popupbackground").hide();
78         $("#popupwrap").hide();
79
80         if (!game.auction()) {
81             play();
82         }
83     };
84
85     this.addPropertyToAuctionQueue = function(propertyIndex) {
86         auctionQueue.push(propertyIndex);
87     };
88
89     this.auction = function() {
90         if (auctionQueue.length === 0) {
91             return false;
92         }
93
94         index = auctionQueue.shift();
95
96         var s = square[index];
97
98         if (s.price === 0 || s.owner !== 0) {
99             return game.auction();
100         }
101
102         auctionproperty = index;
103         highestbidder = 0;
104         highestbid = 0;
105         currentbidder = turn + 1;
106
107         if (currentbidder > pcount) {
108             currentbidder -= pcount;
109         }
110
111         popup("<div style='font-weight: bold; font-size: 16px; margin-bottom: 10px;'>
Auction <span id='propertyname'></span></div><div>Highest Bid = $<span id='highestbid'><
/ span> (<span id='highestbidder'></span></div><div><span id='currentbidder'></span>, it
is your turn to bid.</div><div><input id='bid' title='Enter an amount to bid on " + s.name
+ "." style='width: 291px;' /></div><div><input type='button' value='Bid' onclick='
game.auctionBid();' title='Place your bid.' /><input type='button' value='Pass' title='
Skip bidding this time.' onclick='game.auctionPass();' /><input type='button' value='Exit

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Auction' title='Stop bidding on " + s.name + " altogether.' onclick='if (confirm(\"Are
you sure you want to stop bidding on this property altogether?\")) game.auctionExit();'
/></div>", "blank");
112
113     document.getElementById("propertyname").innerHTML = "<a href='javascript:void(0);'
' onmouseover='showdeed(" + auctionproperty + ");' onmouseout='hidedeed();' class='
statscellcolor'>" + s.name + "</a>";
114     document.getElementById("highestbid").innerHTML = "0";
115     document.getElementById("highestbidder").innerHTML = "N/A";
116     document.getElementById("currentbidder").innerHTML = player[currentbidder].name;
117     document.getElementById("bid").onkeydown = function (e) {
118         var key = 0;
119         var isCtrl = false;
120         var isShift = false;
121
122
123         if (window.event) {
124             key = window.event.keyCode;
125             isCtrl = window.event.ctrlKey;
126             isShift = window.event.shiftKey;
127         } else if (e) {
128             key = e.keyCode;
129             isCtrl = e.ctrlKey;
130             isShift = e.shiftKey;
131         }
132
133         if (isNaN(key)) {
134             return true;
135         }
136
137         if (key === 13) {
138             game.auctionBid();
139             return false;
140         }
141
142         // Allow backspace, tab, delete, arrow keys, or if control was pressed,
respectively.
143         if (key === 8 || key === 9 || key === 46 || (key >= 35 && key <= 40) ||
isCtrl) {
144             return true;
145         }
146
147         if (isShift) {
148             return false;
149         }
150
151         // Only allow number keys.
152         return (key >= 48 && key <= 57) || (key >= 96 && key <= 105);
153     };
154
155     document.getElementById("bid").onfocus = function () {
156         this.style.color = "black";
157         if (isNaN(this.value)) {
158             this.value = "";
159         }
160     };
161
162     updateATP();
163
164     if (!player[currentbidder].human) {
165         currentbidder = turn; // auctionPass advances currentbidder.

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```
166         this.auctionPass();
167     }
168     return true;
169 };
170
171 this.auctionPass = function() {
172     if (highestbidder === 0) {
173         highestbidder = currentbidder;
174     }
175
176     while (true) {
177         currentbidder++;
178
179         if (currentbidder > pcount) {
180             currentbidder -= pcount;
181         }
182
183         if (currentbidder == highestbidder) {
184             finalizeAuction();
185             return;
186         } else if (player[currentbidder].bidding) {
187             var p = player[currentbidder];
188
189             if (!p.human) {
190                 var bid = p.AI.bid(auctionproperty, highestbid);
191
192                 if (bid === -1 || highestbid >= p.money) {
193                     p.bidding = false;
194
195                     window.alert(p.name + " exited the auction.");
196                     continue;
197
198                 } else if (bid === 0) {
199                     window.alert(p.name + " passed.");
200                     continue;
201
202                 } else if (bid > 0) {
203                     this.auctionBid(bid);
204                     window.alert(p.name + " bid ATP" + bid + ".");
205                     continue;
206                 }
207             }
208             return;
209         } else {
210             break;
211         }
212     }
213
214     document.getElementById("currentbidder").innerHTML = player[currentbidder].name;
215     document.getElementById("bid").value = "";
216     document.getElementById("bid").style.color = "black";
217 };
218
219
220 this.auctionBid = function(bid) {
221     bid = bid || parseInt(document.getElementById("bid").value, 10);
222
223     if (bid === "" || bid === null) {
224         document.getElementById("bid").value = "Please enter a bid.";
225         document.getElementById("bid").style.color = "red";
```



```
226     } else if (isNaN(bid)) {
227         document.getElementById("bid").value = "Your bid must be a number.";
228         document.getElementById("bid").style.color = "red";
229     } else {
230
231         if (bid > player[currentbidder].money) {
232             document.getElementById("bid").value = "You don't have enough money to
bid ATP" + bid + ".";
233             document.getElementById("bid").style.color = "red";
234         } else if (bid > highestbid) {
235             highestbid = bid;
236             document.getElementById("highestbid").innerHTML = parseInt(bid, 10);
237             highestbidder = currentbidder;
238             document.getElementById("highestbidder").innerHTML =
player[highestbidder].name;
239
240             document.getElementById("bid").focus();
241
242             if (player[currentbidder].human) {
243                 this.auctionPass();
244             }
245         } else {
246             document.getElementById("bid").value = "Your bid must be greater than
highest bid. (ATP" + highestbid + ")";
247             document.getElementById("bid").style.color = "red";
248         }
249     }
250 };
251
252 this.auctionExit = function() {
253     player[currentbidder].bidding = false;
254     this.auctionPass();
255 };
256
257
258
259 // Trade functions:
260
261
262
263 var currentInitiator;
264 var currentRecipient;
265
266 // Define event handlers:
267
268 var tradeMoneyOnKeyDown = function (e) {
269     var key = 0;
270     var isCtrl = false;
271     var isShift = false;
272
273     if (window.event) {
274         key = window.event.keyCode;
275         isCtrl = window.event.ctrlKey;
276         isShift = window.event.shiftKey;
277     } else if (e) {
278         key = e.keyCode;
279         isCtrl = e.ctrlKey;
280         isShift = e.shiftKey;
281     }
282
283     if (isNaN(key)) {
```

```
284         return true;
285     }
286
287     if (key === 13) {
288         return false;
289     }
290
291     // Allow backspace, tab, delete, arrow keys, or if control was pressed,
    respectively.
292     if (key === 8 || key === 9 || key === 46 || (key >= 35 && key <= 40) || isCtrl) {
293         return true;
294     }
295
296     if (isShift) {
297         return false;
298     }
299
300     // Only allow number keys.
301     return (key >= 48 && key <= 57) || (key >= 96 && key <= 105);
302 };
303
304 var tradeMoneyOnFocus = function () {
305     this.style.color = "black";
306     if (isNaN(this.value) || this.value === "0") {
307         this.value = "";
308     }
309 };
310
311 var tradeMoneyOnChange = function(e) {
312     $("#proposetradebutton").show();
313     $("#canceltradebutton").show();
314     $("#accepttradebutton").hide();
315     $("#rejecttradebutton").hide();
316
317     var amount = this.value;
318
319     if (isNaN(amount)) {
320         this.value = "This value must be a number.";
321         this.style.color = "red";
322         return false;
323     }
324
325     amount = Math.round(amount) || 0;
326     this.value = amount;
327
328     if (amount < 0) {
329         this.value = "This value must be greater than 0.";
330         this.style.color = "red";
331         return false;
332     }
333
334     return true;
335 };
336
337 document.getElementById("trade-lefttp-money").onkeydown = tradeMoneyOnKeyDown;
338 document.getElementById("trade-righttp-money").onkeydown = tradeMoneyOnKeyDown;
339 document.getElementById("trade-lefttp-money").onfocus = tradeMoneyOnFocus;
340 document.getElementById("trade-righttp-money").onfocus = tradeMoneyOnFocus;
341 document.getElementById("trade-lefttp-money").onchange = tradeMoneyOnChange;
342 document.getElementById("trade-righttp-money").onchange = tradeMoneyOnChange;
```

```

343
344     var resetTrade = function(initiator, recipient, allowRecipientToBeChanged) {
345         var currentSquare;
346         var currentTableRow;
347         var currentTableCell;
348         var currentTableCellCheckbox;
349         var nameSelect;
350         var currentOption;
351         var allGroupUnimproved;
352         var currentName;
353
354         var tableRowOnClick = function(e) {
355             var checkboxElement = this.firstChild.firstChild;
356
357             if (checkboxElement !== e.srcElement) {
358                 checkboxElement.checked = !checkboxElement.checked;
359             }
360
361             $("#proposetradebutton").show();
362             $("#canceltradebutton").show();
363             $("#accepttradebutton").hide();
364             $("#rejecttradebutton").hide();
365         };
366
367         var initiatorProperty = document.getElementById("trade-leftp-property");
368         var recipientProperty = document.getElementById("trade-rightp-property");
369
370         currentInitiator = initiator;
371         currentRecipient = recipient;
372
373         // Empty elements.
374         while (initiatorProperty.lastChild) {
375             initiatorProperty.removeChild(initiatorProperty.lastChild);
376         }
377
378         while (recipientProperty.lastChild) {
379             recipientProperty.removeChild(recipientProperty.lastChild);
380         }
381
382         var initiatorSideTable = document.createElement("table");
383         var recipientSideTable = document.createElement("table");
384
385
386         for (var i = 0; i < 34; i++) {
387             currentSquare = square[i];
388
389             // A property cannot be traded if any properties in its group have been
improved.
390             if (currentSquare.house > 0 || currentSquare.groupNumber === 0) {
391                 continue;
392             }
393
394             allGroupUnimproved = true;
395             var max = currentSquare.group.length;
396             for (var j = 0; j < max; j++) {
397
398                 if (square[currentSquare.group[j]].house > 0) {
399                     allGroupUnimproved = false;
400                     break;
401                 }

```

```

402     }
403
404     if (!allGroupUnimproved) {
405         continue;
406     }
407
408     // Offered properties.
409     if (currentSquare.owner === initiator.index) {
410         currentTableRow = initiatorSideTable.appendChild(document.createElement("
411 tr"));
412         currentTableRow.onclick = tableRowOnClick;
413
414         currentTableCell = currentTableRow.appendChild(document.createElement("
415 td"));
416         currentTableCell.className = "propertycellcheckbox";
417         currentTableCellCheckbox =
418 currentTableCell.appendChild(document.createElement("input"));
419         currentTableCellCheckbox.type = "checkbox";
420         currentTableCellCheckbox.id = "tradeleftcheckbox" + i;
421         currentTableCellCheckbox.title = "Check this box to include " +
422 currentSquare.name + " in the trade.";
423
424         currentTableCell = currentTableRow.appendChild(document.createElement("
425 td"));
426         currentTableCell.className = "propertycellcolor";
427         currentTableCell.style.backgroundColor = currentSquare.color;
428
429         if (currentSquare.groupNumber === 0) {
430             currentTableCell.style.borderColor = "grey";
431         } else {
432             currentTableCell.style.borderColor = currentSquare.color;
433         }
434
435         currentTableCell.propertyIndex = i;
436         currentTableCell.onmouseover = function() {showdeed(this.propertyIndex);}
437 ;
438         currentTableCell.onmouseout = hidedeed;
439
440         currentTableCell = currentTableRow.appendChild(document.createElement("
441 td"));
442         currentTableCell.className = "propertycellname";
443         if (currentSquare.mortgage) {
444             currentTableCell.title = "Mortgaged";
445             currentTableCell.style.color = "grey";
446         }
447         currentTableCell.textContent = currentSquare.name;
448
449     // Requested properties.
450     } else if (currentSquare.owner === recipient.index) {
451         currentTableRow = recipientSideTable.appendChild(document.createElement("
452 tr"));
453         currentTableRow.onclick = tableRowOnClick;
454
455         currentTableCell = currentTableRow.appendChild(document.createElement("
456 td"));
457         currentTableCell.className = "propertycellcheckbox";
458         currentTableCellCheckbox =
459 currentTableCell.appendChild(document.createElement("input"));
460         currentTableCellCheckbox.type = "checkbox";
461         currentTableCellCheckbox.id = "traderightcheckbox" + i;
462         currentTableCellCheckbox.title = "Check this box to include " +
463 currentSquare.name + " in the trade.";
464

```

```
454         currentTableCell = currentTableRow.appendChild(document.createElement("
td"));
455         currentTableCell.className = "propertycellcolor";
456         currentTableCell.style.backgroundColor = currentSquare.color;
457
458         if (currentSquare.groupNumber === 0) {
459             currentTableCell.style.borderColor = "grey";
460         } else {
461             currentTableCell.style.borderColor = currentSquare.color;
462         }
463
464         currentTableCell.propertyIndex = i;
465         currentTableCell.onmouseover = function() {showdeed(this.propertyIndex);}
;
466         currentTableCell.onmouseout = hidedeed;
467
468         currentTableCell = currentTableRow.appendChild(document.createElement("
td"));
469         currentTableCell.className = "propertycellname";
470         if (currentSquare.mortgage) {
471             currentTableCell.title = "Mortgaged";
472             currentTableCell.style.color = "grey";
473         }
474         currentTableCell.textContent = currentSquare.name;
475     }
476 }
477
478 if (initiator.specialstationJailCard) {
479     currentTableRow = initiatorSideTable.appendChild(document.createElement("tr")
);
480     currentTableRow.onclick = tableRowOnClick;
481
482     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
483     currentTableCell.className = "propertycellcheckbox";
484     currentTableCellCheckbox =
currentTableCell.appendChild(document.createElement("input"));
485     currentTableCellCheckbox.type = "checkbox";
486     currentTableCellCheckbox.id = "tradeleftcheckbox40";
487     currentTableCellCheckbox.title = "Check this box to include this Get Out of
Enzyme Inhibition Zone Free Card in the trade.";
488
489     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
490     currentTableCell.className = "propertycellcolor";
491     currentTableCell.style.backgroundColor = "white";
492     currentTableCell.style.borderColor = "grey";
493
494     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
495     currentTableCell.className = "propertycellname";
496
497     currentTableCell.textContent = "Get Out of Enzyme Inhibition Zone Free Card";
498 } else if (recipient.specialstationJailCard) {
499     currentTableRow = recipientSideTable.appendChild(document.createElement("tr")
);
500     currentTableRow.onclick = tableRowOnClick;
501
502     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
503     currentTableCell.className = "propertycellcheckbox";
504     currentTableCellCheckbox =
currentTableCell.appendChild(document.createElement("input"));
505     currentTableCellCheckbox.type = "checkbox";
506     currentTableCellCheckbox.id = "traderightcheckbox40";
```

```
507         currentTableCellCheckbox.title = "Check this box to include this Get Out of
Enzyme Inhibition Zone Free Card in the trade.";
508
509         currentTableCell = currentTableRow.appendChild(document.createElement("td"));
510         currentTableCell.className = "propertycellcolor";
511         currentTableCell.style.backgroundColor = "white";
512         currentTableCell.style.borderColor = "grey";
513
514         currentTableCell = currentTableRow.appendChild(document.createElement("td"));
515         currentTableCell.className = "propertycellname";
516
517         currentTableCell.textContent = "Get Out of Enzyme Inhibition Zone Free Card";
518     }
519
520     if (initiator.chanceJailCard) {
521         currentTableRow = initiatorSideTable.appendChild(document.createElement("tr"))
522     };
523
524     currentTableRow.onclick = tableRowOnClick;
525
526     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
527     currentTableCell.className = "propertycellcheckbox";
528     currentTableCellCheckbox =
currentTableCell.appendChild(document.createElement("input"));
529     currentTableCellCheckbox.type = "checkbox";
530     currentTableCellCheckbox.id = "tradeleftcheckbox41";
531     currentTableCellCheckbox.title = "Check this box to include this Get Out of
Enzyme Inhibition Zone Free Card in the trade.";
532
533     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
534     currentTableCell.className = "propertycellcolor";
535     currentTableCell.style.backgroundColor = "white";
536     currentTableCell.style.borderColor = "grey";
537
538     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
539     currentTableCell.className = "propertycellname";
540
541     currentTableCell.textContent = "Get Out of Enzyme Inhibition Zone Free Card";
542 } else if (recipient.chanceJailCard) {
543     currentTableRow = recipientSideTable.appendChild(document.createElement("tr"))
544 };
545
546     currentTableRow.onclick = tableRowOnClick;
547
548     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
549     currentTableCell.className = "propertycellcheckbox";
550     currentTableCellCheckbox =
currentTableCell.appendChild(document.createElement("input"));
551     currentTableCellCheckbox.type = "checkbox";
552     currentTableCellCheckbox.id = "traderightcheckbox41";
553     currentTableCellCheckbox.title = "Check this box to include this Get Out of
Enzyme Inhibition Zone Free Card in the trade.";
554
555     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
556     currentTableCell.className = "propertycellcolor";
557     currentTableCell.style.backgroundColor = "white";
558     currentTableCell.style.borderColor = "grey";
559
560     currentTableCell = currentTableRow.appendChild(document.createElement("td"));
561     currentTableCell.className = "propertycellname";
562
563     currentTableCell.textContent = "Get Out of Enzyme Inhibition Zone Free Card";
564 }
```

```
562     if (initiatorSideTable.lastChild) {
563         initiatorProperty.appendChild(initiatorSideTable);
564     } else {
565         initiatorProperty.textContent = initiator.name + " has no properties to
trade.";
566     }
567
568     if (recipientSideTable.lastChild) {
569         recipientProperty.appendChild(recipientSideTable);
570     } else {
571         recipientProperty.textContent = recipient.name + " has no properties to
trade.";
572     }
573
574     document.getElementById("trade-leftp-name").textContent = initiator.name;
575
576     currentName = document.getElementById("trade-rightp-name");
577
578     if (allowRecipientToBeChanged && pcount > 2) {
579         // Empty element.
580         while (currentName.lastChild) {
581             currentName.removeChild(currentName.lastChild);
582         }
583
584         nameSelect = currentName.appendChild(document.createElement("select"));
585         for (var i = 1; i <= pcount; i++) {
586             if (i === initiator.index) {
587                 continue;
588             }
589
590             currentOption = nameSelect.appendChild(document.createElement("option"));
591             currentOption.value = i + "";
592             currentOption.style.color = player[i].color;
593             currentOption.textContent = player[i].name;
594
595             if (i === recipient.index) {
596                 currentOption.selected = "selected";
597             }
598         }
599
600         nameSelect.onchange = function() {
601             resetTrade(currentInitiator, player[parseInt(this.value, 10)], true);
602         };
603
604         nameSelect.title = "Select a player to trade with.";
605     } else {
606         currentName.textContent = recipient.name;
607     }
608
609     document.getElementById("trade-leftp-money").value = "0";
610     document.getElementById("trade-rightp-money").value = "0";
611
612 };
613
614 var readTrade = function() {
615     var initiator = currentInitiator;
616     var recipient = currentRecipient;
617     var property = new Array(40);
618     var money;
619     var specialstationJailCard;
```



```
620     var chanceJailCard;
621
622     for (var i = 0; i < 34; i++) {
623
624         if (document.getElementById("tradeleftcheckbox" + i) &&
document.getElementById("tradeleftcheckbox" + i).checked) {
625             property[i] = 1;
626         } else if (document.getElementById("traderightcheckbox" + i) &&
document.getElementById("traderightcheckbox" + i).checked) {
627             property[i] = -1;
628         } else {
629             property[i] = 0;
630         }
631     }
632
633     if (document.getElementById("tradeleftcheckbox40") && document.getElementById("
tradeleftcheckbox40").checked) {
634         specialstationJailCard = 1;
635     } else if (document.getElementById("traderightcheckbox40") &&
document.getElementById("traderightcheckbox40").checked) {
636         specialstationJailCard = -1;
637     } else {
638         specialstationJailCard = 0;
639     }
640
641     if (document.getElementById("tradeleftcheckbox41") && document.getElementById("
tradeleftcheckbox41").checked) {
642         chanceJailCard = 1;
643     } else if (document.getElementById("traderightcheckbox41") &&
document.getElementById("traderightcheckbox41").checked) {
644         chanceJailCard = -1;
645     } else {
646         chanceJailCard = 0;
647     }
648
649     money = parseInt(document.getElementById("trade-leftp-money").value, 10) || 0;
650     money -= parseInt(document.getElementById("trade-rightp-money").value, 10) || 0;
651
652     var trade = new Trade(initiator, recipient, money, property,
specialstationJailCard, chanceJailCard);
653
654     return trade;
655 };
656
657 var writeTrade = function(tradeObj) {
658     resetTrade(tradeObj.getInitiator(), tradeObj.getRecipient(), false);
659
660     for (var i = 0; i < 34; i++) {
661
662         if (document.getElementById("tradeleftcheckbox" + i)) {
663             document.getElementById("tradeleftcheckbox" + i).checked = false;
664             if (tradeObj.getProperty(i) === 1) {
665                 document.getElementById("tradeleftcheckbox" + i).checked = true;
666             }
667         }
668
669         if (document.getElementById("traderightcheckbox" + i)) {
670             document.getElementById("traderightcheckbox" + i).checked = false;
671             if (tradeObj.getProperty(i) === -1) {
672                 document.getElementById("traderightcheckbox" + i).checked = true;
673             }
674         }
675     }
676 }
```



```

675     }
676
677     if (document.getElementById("tradeleftcheckbox40")) {
678         if (tradeObj.getspecialstationJailCard() === 1) {
679             document.getElementById("tradeleftcheckbox40").checked = true;
680         } else {
681             document.getElementById("tradeleftcheckbox40").checked = false;
682         }
683     }
684
685     if (document.getElementById("traderightcheckbox40")) {
686         if (tradeObj.getspecialstationJailCard() === -1) {
687             document.getElementById("traderightcheckbox40").checked = true;
688         } else {
689             document.getElementById("traderightcheckbox40").checked = false;
690         }
691     }
692
693     if (document.getElementById("tradeleftcheckbox41")) {
694         if (tradeObj.getChanceJailCard() === 1) {
695             document.getElementById("tradeleftcheckbox41").checked = true;
696         } else {
697             document.getElementById("tradeleftcheckbox41").checked = false;
698         }
699     }
700
701     if (document.getElementById("traderightcheckbox41")) {
702         if (tradeObj.getChanceJailCard() === -1) {
703             document.getElementById("traderightcheckbox41").checked = true;
704         } else {
705             document.getElementById("traderightcheckbox41").checked = false;
706         }
707     }
708
709     if (tradeObj.getMoney() > 0) {
710         document.getElementById("trade-lefttp-money").value = tradeObj.getMoney() + ""
711     }
712     } else {
713         document.getElementById("trade-righttp-money").value = (-tradeObj.getMoney())
714         + "";
715     }
716
717     };
718
719     this.trade = function(tradeObj) {
720         $("#board").hide();
721         $("#control").hide();
722         $("#trade").show();
723         $("#proposetradebutton").show();
724         $("#canceltradebutton").show();
725         $("#accepttradebutton").hide();
726         $("#rejecttradebutton").hide();
727
728         if (tradeObj instanceof Trade) {
729             writeTrade(tradeObj);
730             this.proposeTrade();
731         } else {
732             var initiator = player[turn];
733             var recipient = turn === 1 ? player[2] : player[1];

```

```
733         currentInitiator = initiator;
734         currentRecipient = recipient;
735
736         resetTrade(initiator, recipient, true);
737     }
738 };
739
740
741 this.cancelTrade = function() {
742     $("#board").show();
743     $("#control").show();
744     $("#trade").hide();
745
746
747     if (!player[turn].human) {
748         player[turn].AI.alertList = "";
749         game.next();
750     }
751
752 };
753
754 this.acceptTrade = function(tradeObj) {
755     if (isNaN(document.getElementById("trade-leftp-money").value)) {
756 number."";
757         document.getElementById("trade-leftp-money").value = "This value must be a
758         document.getElementById("trade-leftp-money").style.color = "red";
759         return false;
760     }
761
762     if (isNaN(document.getElementById("trade-rightp-money").value)) {
763 number."";
764         document.getElementById("trade-rightp-money").value = "This value must be a
765         document.getElementById("trade-rightp-money").style.color = "red";
766         return false;
767     }
768
769     var showAlerts = true;
770     var money;
771     var initiator;
772     var recipient;
773
774     if (tradeObj) {
775         showAlerts = false;
776     } else {
777         tradeObj = readTrade();
778     }
779
780     money = tradeObj.getMoney();
781     initiator = tradeObj.getInitiator();
782     recipient = tradeObj.getRecipient();
783
784     if (money > 0 && money > initiator.money) {
785         document.getElementById("trade-leftp-money").value = initiator.name + " does
786 not have ATP" + money + ".";
787         document.getElementById("trade-leftp-money").style.color = "red";
788         return false;
789     } else if (money < 0 && -money > recipient.money) {
790         document.getElementById("trade-rightp-money").value = recipient.name + " does
791 not have ATP" + (-money) + ".";
792         document.getElementById("trade-rightp-money").style.color = "red";
793     }
794 }
```

```
790         return false;
791     }
792
793     var isAPropertySelected = 0;
794
795     // Ensure that some properties are selected.
796     for (var i = 0; i < 34; i++) {
797         isAPropertySelected |= tradeObj.getProperty(i);
798     }
799
800     isAPropertySelected |= tradeObj.getspecialstationJailCard();
801     isAPropertySelected |= tradeObj.getChanceJailCard();
802
803     if (isAPropertySelected === 0) {
804         popup("<p>One or more properties must be selected in order to trade.</p>");
805
806         return false;
807     }
808
809     if (showAlerts && !confirm(initiator.name + ", are you sure you want to make this
exchange with " + recipient.name + "?")) {
810         return false;
811     }
812
813     // Exchange properties
814     for (var i = 0; i < 34; i++) {
815
816         if (tradeObj.getProperty(i) === 1) {
817             square[i].owner = recipient.index;
818             addAlert(recipient.name + " received " + square[i].name + " from " +
initiator.name + ".");
819         } else if (tradeObj.getProperty(i) === -1) {
820             square[i].owner = initiator.index;
821             addAlert(initiator.name + " received " + square[i].name + " from " +
recipient.name + ".");
822         }
823     }
824
825
826     if (tradeObj.getspecialstationJailCard() === 1) {
827         initiator.specialstationJailCard = false;
828         recipient.specialstationJailCard = true;
829         addAlert(recipient.name + ' received a "Get Out of Enzyme Inhibition Zone
Free" card from ' + initiator.name + ".");
830     } else if (tradeObj.getspecialstationJailCard() === -1) {
831         initiator.specialstationJailCard = true;
832         recipient.specialstationJailCard = false;
833         addAlert(initiator.name + ' received a "Get Out of Enzyme Inhibition Zone
Free" card from ' + recipient.name + ".");
834     }
835
836     if (tradeObj.getChanceJailCard() === 1) {
837         initiator.chanceJailCard = false;
838         recipient.chanceJailCard = true;
839         addAlert(recipient.name + ' received a "Get Out of Enzyme Inhibition Zone
Free" card from ' + initiator.name + ".");
840     } else if (tradeObj.getChanceJailCard() === -1) {
841         initiator.chanceJailCard = true;
842         recipient.chanceJailCard = false;
843         addAlert(initiator.name + ' received a "Get Out of Enzyme Inhibition Zone
Free" card from ' + recipient.name + ".");
844     }
```

```

845
846 // Exchange money.
847 if (money > 0) {
848     initiator.pay(money, recipient.index);
849     recipient.money += money;
850
851     addAlert(recipient.name + " received ATP" + money + " from " + initiator.name
+ ".");
852 } else if (money < 0) {
853     money = -money;
854
855     recipient.pay(money, initiator.index);
856     initiator.money += money;
857
858     addAlert(initiator.name + " received ATP" + money + " from " + recipient.name
+ ".");
859 }
860
861 updateOwned();
862 updateATP();
863
864 $("#board").show();
865 $("#control").show();
866 $("#trade").hide();
867
868 if (!player[turn].human) {
869     player[turn].AI.alertList = "";
870     game.next();
871 }
872 };
873
874 this.proposeTrade = function() {
875     if (isNaN(document.getElementById("trade-leftp-money").value)) {
876         document.getElementById("trade-leftp-money").value = "This value must be a
number.";
877         document.getElementById("trade-leftp-money").style.color = "red";
878         return false;
879     }
880
881     if (isNaN(document.getElementById("trade-rightp-money").value)) {
882         document.getElementById("trade-rightp-money").value = "This value must be a
number.";
883         document.getElementById("trade-rightp-money").style.color = "red";
884         return false;
885     }
886
887     var tradeObj = readTrade();
888     var money = tradeObj.getMoney();
889     var initiator = tradeObj.getInitiator();
890     var recipient = tradeObj.getRecipient();
891     var reversedTradeProperty = [];
892
893     if (money > 0 && money > initiator.money) {
894         document.getElementById("trade-leftp-money").value = initiator.name + " does
not have ATP" + money + ".";
895         document.getElementById("trade-leftp-money").style.color = "red";
896         return false;
897     } else if (money < 0 && -money > recipient.money) {
898         document.getElementById("trade-rightp-money").value = recipient.name + " does
not have ATP" + (-money) + ".";
899         document.getElementById("trade-rightp-money").style.color = "red";

```

```
900         return false;
901     }
902
903     var isAPropertySelected = 0;
904
905     // Ensure that some properties are selected.
906     for (var i = 0; i < 34; i++) {
907         reversedTradeProperty[i] = -tradeObj.getProperty(i);
908         isAPropertySelected |= tradeObj.getProperty(i);
909     }
910
911     isAPropertySelected |= tradeObj.getspecialstationJailCard();
912     isAPropertySelected |= tradeObj.getChanceJailCard();
913
914     if (isAPropertySelected === 0) {
915         popup("<p>One or more properties must be selected in order to trade.</p>");
916
917         return false;
918     }
919
920     if (initiator.human && !confirm(initiator.name + ", are you sure you want to make
this offer to " + recipient.name + "?")) {
921         return false;
922     }
923
924     var reversedTrade = new Trade(recipient, initiator, -money,
reversedTradeProperty, -tradeObj.getspecialstationJailCard(), -
tradeObj.getChanceJailCard());
925
926     if (recipient.human) {
927
928         writeTrade(reversedTrade);
929
930         $("#proposetradebutton").hide();
931         $("#canceltradebutton").hide();
932         $("#accepttradebutton").show();
933         $("#rejecttradebutton").show();
934
935         addAlert(initiator.name + " initiated a trade with " + recipient.name + ".");
936         popup("<p>" + initiator.name + " has proposed a trade with you, " +
recipient.name + ". You may accept, reject, or modify the offer.</p>");
937     } else {
938         var tradeResponse = recipient.AI.acceptTrade(tradeObj);
939
940         if (tradeResponse === true) {
941             popup("<p>" + recipient.name + " has accepted your offer.</p>");
942             this.acceptTrade(reversedTrade);
943         } else if (tradeResponse === false) {
944             popup("<p>" + recipient.name + " has declined your offer.</p>");
945             return;
946         } else if (tradeResponse instanceof Trade) {
947             popup("<p>" + recipient.name + " has proposed a counteroffer.</p>");
948             writeTrade(tradeResponse);
949
950             $("#proposetradebutton, #canceltradebutton").hide();
951             $("#accepttradebutton").show();
952             $("#rejecttradebutton").show();
953         }
954     }
955 };
956
```

```

957
958
959 // Bankruptcy functions:
960
961
962
963
964 this.eliminatePlayer = function() {
965     var p = player[turn];
966
967     for (var i = p.index; i < pcount; i++) {
968         player[i] = player[i + 1];
969         player[i].index = i;
970     }
971
972
973     for (var i = 0; i < 34; i++) {
974         if (square[i].owner >= p.index) {
975             square[i].owner--;
976         }
977     }
978
979     pcount--;
980     turn--;
981
982     if (pcount === 2) {
983         document.getElementById("stats").style.width = "454px";
984     } else if (pcount === 3) {
985         document.getElementById("stats").style.width = "686px";
986     }
987
988     if (pcount === 1) {
989         updateATP();
990         $("#control").hide();
991         $("#board").hide();
992         $("#refresh").show();
993
994         // // Display land counts for survey purposes.
995         // var text;
996         // for (var i = 0; i < 34; i++) {
997             // if (i === 0)
998                 // text = square[i].landcount;
999             // else
1000                 // text += " " + square[i].landcount;
1001         // }
1002         // document.getElementById("refresh").innerHTML += "<br><br><div><textarea
type='text' style='width: 980px;' onclick='javascript:select();' />" + text + "<
/textarea></div>";
1003
1004         popup("<p>Congratulations, " + player[1].name + ", you have won the game.</p>
<div>");
1005
1006     } else {
1007         play();
1008     }
1009 };
1010
1011 this.bankruptcyUnmortgage = function() {
1012     var p = player[turn];
1013
1014     if (p.creditor === 0) {

```

```

1015         game.eliminatePlayer();
1016         return;
1017     }
1018
1019     var HTML = "<p>" + player[p.creditor].name + ", you may unmortgage any of the
following properties, interest free, by clicking on them. Click OK when finished.</p><
table>";
1020     var price;
1021
1022     for (var i = 0; i < 34; i++) {
1023         sq = square[i];
1024         if (sq.owner == p.index && sq.mortgage) {
1025             price = Math.round(sq.price * 0.5);
1026
1027             HTML += "<tr><td class='propertycellcolor' style='background: " +
sq.color + "';";
1028
1029             if (sq.groupNumber == 1 || sq.groupNumber == 2) {
1030                 HTML += " border: 1px solid grey;";
1031             } else {
1032                 HTML += " border: 1px solid " + sq.color + "';";
1033             }
1034
1035             // Player already paid interest, so they can unmortgage for the mortgage
price.
1036             HTML += "' onmouseover='showdeed(" + i + ");' onmouseout='hidedeed();'><
/td><td class='propertycellname'><a href='javascript:void(0);' title='Unmortgage " +
sq.name + " for $" + price + ".'" onclick='if (" + price + " <= player[" + p.creditor + "]
.money) {player[" + p.creditor + "].pay(" + price + ", 0); square[" + i + "].mortgage =
false; addAlert(\"" + player[p.creditor].name + " unmortgaged " + sq.name + " for $" +
price + ".\");} this.parentElement.parentElement.style.display = \"none\";'>Unmortgage "
+ sq.name + " ($" + price + ")</a></td></tr>";
1037
1038             sq.owner = p.creditor;
1039
1040         }
1041     }
1042
1043     HTML += "</table>";
1044
1045     popup(HTML, game.eliminatePlayer);
1046 };
1047
1048 this.resign = function() {
1049     popup("<p>Are you sure you want to resign?</p>", game.bankruptcy, "Yes/No");
1050 };
1051
1052 this.bankruptcy = function() {
1053     var p = player[turn];
1054     var pcredit = player[p.creditor];
1055     var bankruptcyUnmortgageFee = 0;
1056
1057     if (p.money >= 0) {
1058         return;
1059     }
1060
1061     addAlert(p.name + " is bankrupt.");
1062
1063     if (p.creditor !== 0) {
1064         pcredit.money += p.money;
1065     }
1066

```

```
1067
1068     for (var i = 0; i < 34; i++) {
1069         sq = square[i];
1070         if (sq.owner == p.index) {
1071             // Mortgaged properties will be tranfered by bankruptcyUnmortgage();
1072             if (!sq.mortgage) {
1073                 sq.owner = p.creditor;
1074             } else {
1075                 bankruptcyUnmortgageFee += Math.round(sq.price * 0.1);
1076             }
1077
1078             if (sq.house > 0) {
1079                 if (p.creditor !== 0) {
1080                     pcredit.money += sq.houseprice * 0.5 * sq.house;
1081                 }
1082                 sq.hotel = 0;
1083                 sq.house = 0;
1084             }
1085
1086             if (p.creditor === 0) {
1087                 sq.mortgage = false;
1088                 game.addPropertyToAuctionQueue(i);
1089                 sq.owner = 0;
1090             }
1091         }
1092     }
1093
1094     updateATP();
1095
1096     if (p.chanceJailCard) {
1097         p.chanceJailCard = false;
1098         pcredit.chanceJailCard = true;
1099     }
1100
1101     if (p.specialstationJailCard) {
1102         p.specialstationJailCard = false;
1103         pcredit.specialstationJailCard = true;
1104     }
1105
1106     if (pcount === 2 || bankruptcyUnmortgageFee === 0 || p.creditor === 0) {
1107         game.eliminatePlayer();
1108     } else {
1109         addAlert(pcredit.name + " paid ATP" + bankruptcyUnmortgageFee + " interest on
the mortgaged properties received from " + p.name + ".");
1110         popup("<p>" + pcredit.name + ", you must pay ATP" + bankruptcyUnmortgageFee +
" interest on the mortgaged properties you received from " + p.name + ".</p>", function()
{player[pcredit.index].pay(bankruptcyUnmortgageFee, 0); game.bankruptcyUnmortgage();});
1111     }
1112 };
1113 updateBoardUI();
1114 }
1115
1116 var game;
1117
1118
1119 function Player(name, color) {
1120     this.name = name;
1121     this.color = color;
1122     this.position = 0;
1123     this.money = 1500;
1124     this.creditor = -1;
```



```
1125     this.jail = false;
1126     this.jailroll = 0;
1127     this.specialstationJailCard = false;
1128     this.chanceJailCard = false;
1129     this.bidding = true;
1130     this.human = true;
1131     // this.AI = null;
1132
1133     this.pay = function (amount, creditor) {
1134         if (amount <= this.money) {
1135             this.money -= amount;
1136
1137             updateATP();
1138
1139             return true;
1140         } else {
1141             this.money -= amount;
1142             this.creditor = creditor;
1143
1144             updateATP();
1145
1146             return false;
1147         }
1148     };
1149 }
1150
1151 // paramaters:
1152 // initiator: object Player
1153 // recipient: object Player
1154 // money: integer, positive for offered, negative for requested
1155 // property: array of integers, length: 40
1156 // specialstationJailCard: integer, 1 means offered, -1 means requested, 0 means neither
1157 // chanceJailCard: integer, 1 means offered, -1 means requested, 0 means neither
1158 function Trade(initiator, recipient, money, property, specialstationJailCard,
1159 chanceJailCard) {
1160     // For each property and get out of Enzyme Inhibition Zone free cards, 1 means
1161     // offered, -1 means requested, 0 means neither.
1162
1163     this.getInitiator = function() {
1164         return initiator;
1165     };
1166
1167     this.getRecipient = function() {
1168         return recipient;
1169     };
1170
1171     this.getProperty = function(index) {
1172         return property[index];
1173     };
1174
1175     this.getMoney = function() {
1176         return money;
1177     };
1178
1179     this.getspecialstationJailCard = function() {
1180         return specialstationJailCard;
1181     };
1182
1183     this.getChanceJailCard = function() {
1184         return chanceJailCard;
1185     };
1186 }
```

```
1183     };
1184 }
1185
1186 var player = [];
1187 var pcount;
1188 var turn = 0, doublecount = 0;
1189 // Overwrite an array with numbers from one to the array's length in a random order.
1190 Array.prototype.randomize = function(length) {
1191     length = (length || this.length);
1192     var num;
1193     var indexArray = [];
1194
1195     for (var i = 0; i < length; i++) {
1196         indexArray[i] = i;
1197     }
1198
1199     for (var i = 0; i < length; i++) {
1200         // Generate random number between 0 and indexArray.length - 1.
1201         num = Math.floor(Math.random() * indexArray.length);
1202         this[i] = indexArray[num] + 1;
1203
1204         indexArray.splice(num, 1);
1205     }
1206 };
1207
1208 // function show(element) {
1209     // // Element may be an HTML element or the id of one passed as a string.
1210     // if (element.constructor == String) {
1211         // element = document.getElementById(element);
1212     // }
1213
1214     // if (element.tagName == "INPUT" || element.tagName == "SPAN" || element.tagName ==
1215     "LABEL") {
1216         // element.style.display = "inline";
1217     // } else {
1218         // element.style.display = "block";
1219     // }
1220 // }
1221
1222 // function hide(element) {
1223     // // Element may be an HTML element or the id of one passed as a string.
1224     // if (element.constructor == String) {
1225         // document.getElementById(element).style.display = "none";
1226     // } else {
1227         // element.style.display = "none";
1228     // }
1229 // }
1230
1231 function addAlert(alertText) {
1232     $alert = $("#alert");
1233
1234     $(document.createElement("div")).text(alertText).appendTo($alert);
1235
1236     // Animate scrolling down alert element.
1237     $alert.stop().animate({"scrollTop": $alert.prop("scrollHeight")}, 1000);
1238
1239     if (!player[turn].human) {
1240         player[turn].AI.alertList += "<div> " + alertText + "</div>";
1241     }
```

```

1242
1243 function popup(HTML, action, option) {
1244     document.getElementById("popuptext").innerHTML = HTML;
1245     document.getElementById("popup").style.width = "300px";
1246     document.getElementById("popup").style.top = "0px";
1247     document.getElementById("popup").style.left = "0px";
1248
1249     if (!option && typeof action === "string") {
1250         option = action;
1251     }
1252
1253     option = option ? option.toLowerCase() : "";
1254
1255     if (typeof action !== "function") {
1256         action = null;
1257     }
1258
1259     // Yes/No
1260     if (option === "yes/no") {
1261         document.getElementById("popuptext").innerHTML += "<div><input type=\"button\"
value=\"Yes\" id=\"popupy\" /><input type=\"button\" value=\"No\" id=\"popupno\" /><
/div>";
1262
1263         $("#popupy, #popupno").on("click", function() {
1264             $("#popupwrap").hide();
1265             $("#popupbackground").fadeOut(400);
1266         });
1267
1268         $("#popupy").on("click", action);
1269
1270         // Ok
1271     } else if (option !== "blank") {
1272         $("#popuptext").append("<div><input type='button' value='OK' id='popupclose' /><
/div>");
1273         $("#popupclose").focus();
1274
1275         $("#popupclose").on("click", function() {
1276             $("#popupwrap").hide();
1277             $("#popupbackground").fadeOut(400);
1278         }).on("click", action);
1279     }
1280
1281     // Show using animation.
1282     $("#popupbackground").fadeIn(400, function() {
1283         $("#popupwrap").show();
1284     });
1285 }
1286
1287
1288
1289
1290 function updatePosition() {
1291     // Reset borders
1292
1293     for (var i = 0; i < 34; i++) {
1294         document.getElementById("cell" + i).style.border = "1px solid #7c5c36";
1295         document.getElementById("cell" + i + "positionholder").innerHTML = "";
1296     }
1297
1298     var sq, left, top;

```

```
1300
1301     for (var x = 0; x < 34; x++) {
1302         sq = square[x];
1303         left = 0;
1304         top = 0;
1305
1306         for (var y = turn; y <= pcount; y++) {
1307
1308             if (player[y].position == x && !player[y].jail) {
1309
1310                 document.getElementById("cell" + x + "positionholder").innerHTML += "<div
class='cell-position' title='" + player[y].name + "' style='background-color: " +
player[y].color + "; left: " + left + "px; top: " + top + "px;'></div>";
1311                 if (left == 34) {
1312                     left = 0;
1313                     top = 12;
1314                 } else
1315                     left += 12;
1316             }
1317         }
1318
1319         for (var y = 1; y < turn; y++) {
1320
1321             if (player[y].position == x && !player[y].jail) {
1322                 document.getElementById("cell" + x + "positionholder").innerHTML += "<div
class='cell-position' title='" + player[y].name + "' style='background-color: " +
player[y].color + "; left: " + left + "px; top: " + top + "px;'></div>";
1323                 if (left == 34) {
1324                     left = 0;
1325                     top = 12;
1326                 } else
1327                     left += 12;
1328             }
1329         }
1330     }
1331
1332     left = 0;
1333     top = 53;
1334     for (var i = turn; i <= pcount; i++) {
1335         if (player[i].jail) {
1336             document.getElementById("jailpositionholder").innerHTML += "<div class='cell-
position' title='" + player[i].name + "' style='background-color: " + player[i].color + "
; left: " + left + "px; top: " + top + "px;'></div>";
1337             if (left == 34) {
1338                 left = 0;
1339                 top = 41;
1340             } else {
1341                 left += 12;
1342             }
1343         }
1344     }
1345
1346     for (var i = 1; i < turn; i++) {
1347         if (player[i].jail) {
1348             document.getElementById("jailpositionholder").innerHTML += "<div class='cell-
position' title='" + player[i].name + "' style='background-color: " + player[i].color + "
; left: " + left + "px; top: " + top + "px;'></div>";
1349             if (left == 34) {
1350                 left = 0;
1351                 top = 41;
1352             } else
1353                 left += 12;
```

```
1354     }
1355
1356     p = player[turn];
1357
1358     if (p.jail) {
1359         document.getElementById("jail").style.border = "1px solid " + p.color;
1360     } else {
1361         document.getElementById("cell" + p.position).style.border = "1px solid " +
1362         p.color;
1363     }
1364
1365     // for (var i=1; i <= pcount; i++) {
1366     //     document.getElementById("enlarge"+player[i].position+"token").innerHTML+="
```

```
1412     element0.classList.remove("die-no-img");
1413     element1.classList.remove("die-no-img");
1414
1415     element0.title = "Die (" + die0 + " spots)";
1416     element1.title = "Die (" + die1 + " spots)";
1417
1418     if (element0.firstChild) {
1419         element0 = element0.firstChild;
1420     } else {
1421         element0 = element0.appendChild(document.createElement("img"));
1422     }
1423
1424     element0.src = "images/Die_" + die0 + ".png";
1425     element0.alt = die0;
1426
1427     if (element1.firstChild) {
1428         element1 = element1.firstChild;
1429     } else {
1430         element1 = element1.appendChild(document.createElement("img"));
1431     }
1432
1433     element1.src = "images/Die_" + die1 + ".png";
1434     element1.alt = die0;
1435 } else {
1436     document.getElementById("die0").textContent = die0;
1437     document.getElementById("die1").textContent = die1;
1438
1439     document.getElementById("die0").title = "Die";
1440     document.getElementById("die1").title = "Die";
1441 }
1442 }
1443
1444 function updateOwned() {
1445     var p = player[turn];
1446     var checkedproperty = getCheckedProperty();
1447     $("#option").show();
1448     $("#owned").show();
1449
1450     var HTML = "",
1451     firstproperty = -1;
1452
1453     var mortgagetext = "",
1454     housetext = "";
1455     var sq;
1456
1457     for (var i = 0; i < 34; i++) {
1458         sq = square[i];
1459         if (sq.groupNumber && sq.owner === 0) {
1460             $("#cell" + i + "owner").hide();
1461         } else if (sq.groupNumber && sq.owner > 0) {
1462             var currentCellOwner = document.getElementById("cell" + i + "owner");
1463
1464             currentCellOwner.style.display = "block";
1465             currentCellOwner.style.backgroundColor = player[sq.owner].color;
1466             currentCellOwner.title = player[sq.owner].name;
1467         }
1468     }
1469
1470     for (var i = 0; i < 34; i++) {
1471         sq = square[i];
```

```

1472     if (sq.owner == turn) {
1473
1474         mortgagetext = "";
1475         if (sq.mortgage) {
1476             mortgagetext = "title='Mortgaged' style='color: grey;";
1477         }
1478
1479         housetext = "";
1480
1481
1482         if (HTML === "") {
1483             HTML += "<table>";
1484             firstproperty = i;
1485         }
1486
1487         HTML += "<tr class='property-cell-row'><td class='propertycellcheckbox'><
input type='checkbox' id='propertycheckbox" + i + "' /></td><td class='propertycellcolor'
style='background: " + sq.color + ";";
1488
1489         if (sq.groupNumber == 1 || sq.groupNumber == 2) {
1490             HTML += " border: 1px solid grey; width: 18px;";
1491         }
1492
1493         HTML += "' onmouseover='showdeed(" + i + ");' onmouseout='hidedeed();'></td><
td class='propertycellname' " + mortgagetext + ">" + sq.name + housetext + "</td></tr>";
1494     }
1495 }
1496
1497 if (p.specialstationJailCard) {
1498     if (HTML === "") {
1499         firstproperty = 40;
1500         HTML += "<table>";
1501     }
1502     HTML += "<tr class='property-cell-row'><td class='propertycellcheckbox'><input
type='checkbox' id='propertycheckbox40' /></td><td class='propertycellcolor' style='
background: white;'></td><td class='propertycellname'>Get Out of Enzyme Inhibition Zone
Free Card</td></tr>";
1503
1504 }
1505 if (p.chanceJailCard) {
1506     if (HTML === "") {
1507         firstproperty = 41;
1508         HTML += "<table>";
1509     }
1510     HTML += "<tr class='property-cell-row'><td class='propertycellcheckbox'><input
type='checkbox' id='propertycheckbox41' /></td><td class='propertycellcolor' style='
background: white;'></td><td class='propertycellname'>Get Out of Enzyme Inhibition Zone
Free Card</td></tr>";
1511 }
1512
1513 if (HTML === "") {
1514     HTML = p.name + ", you don't have any properties.";
1515     $("#option").hide();
1516 } else {
1517     HTML += "</table>";
1518 }
1519
1520 document.getElementById("owned").innerHTML = HTML;
1521
1522 // Select previously selected property.
1523 if (checkedproperty > -1 && document.getElementById("propertycheckbox" +
checkedproperty)) {

```

```
1524     document.getElementById("propertycheckbox" + checkedproperty).checked = true;
1525 } else if (firstproperty > -1) {
1526     document.getElementById("propertycheckbox" + firstproperty).checked = true;
1527 }
1528 $(".property-cell-row").click(function() {
1529     var row = this;
1530
1531     // Toggle check the current checkbox.
1532     $(this).find(".propertycellcheckbox > input").prop("checked", function(index,
val) {
1533         return !val;
1534     });
1535
1536     // Set all other checkboxes to false.
1537     $(".propertycellcheckbox > input").prop("checked", function(index, val) {
1538         if (!$.contains(row, this)) {
1539             return false;
1540         }
1541     });
1542
1543     updateOption();
1544 });
1545 updateOption();
1546 }
1547
1548 function updateOption() {
1549     $("#option").show();
1550
1551     var allGroupUninproved = true;
1552     var allGroupUnmortgaged = true;
1553     var checkedproperty = getCheckedProperty();
1554
1555     if (checkedproperty < 0 || checkedproperty >= 40) {
1556         $("#buyhousebutton").hide();
1557         $("#sellhousebutton").hide();
1558         $("#mortgagebutton").hide();
1559
1560
1561         var housesum = 32;
1562         var hotelsun = 12;
1563
1564         for (var i = 0; i < 34; i++) {
1565             s = square[i];
1566             if (s.hotel == 1)
1567                 hotelsun--;
1568             else
1569                 housesum -= s.house;
1570         }
1571
1572         $("#buildings").show();
1573
1574         return;
1575     }
1576
1577     $("#buildings").hide();
1578     var sq = square[checkedproperty];
1579
1580     buyhousebutton = document.getElementById("buyhousebutton");
1581     sellhousebutton = document.getElementById("sellhousebutton");
1582
```



```
1583     $("#mortgagebutton").show();
1584     document.getElementById("mortgagebutton").disabled = false;
1585
1586     if (sq.mortgage) {
1587         document.getElementById("mortgagebutton").value = "Unmortgage (ATP" +
1588 Math.round(sq.price * 0.55) + ")";
1589         document.getElementById("mortgagebutton").title = "Unmortgage " + sq.name + " for
1590 ATP" + Math.round(sq.price * 0.55) + ".";
1591         $("#buyhousebutton").hide();
1592         $("#sellhousebutton").hide();
1593
1594         allGroupUnmortgaged = false;
1595     } else {
1596         document.getElementById("mortgagebutton").value = "Mortgage (ATP" + (sq.price *
1597 0.5) + ")";
1598         document.getElementById("mortgagebutton").title = "Mortgage " + sq.name + " for
1599 ATP" + (sq.price * 0.5) + ".";
1600
1601         if (sq.groupNumber >= 3) {
1602             $("#buyhousebutton").show();
1603             $("#sellhousebutton").show();
1604             buyhousebutton.disabled = false;
1605             sellhousebutton.disabled = false;
1606
1607             buyhousebutton.value = "Buy house (ATP" + sq.houseprice + ")";
1608             sellhousebutton.value = "Sell house (ATP" + (sq.houseprice * 0.5) + ")";
1609             buyhousebutton.title = "Buy a house for ATP" + sq.houseprice;
1610             sellhousebutton.title = "Sell a house for ATP" + (sq.houseprice * 0.5);
1611
1612             if (sq.house == 4) {
1613                 buyhousebutton.value = "Buy hotel (ATP" + sq.houseprice + ")";
1614                 buyhousebutton.title = "Buy a hotel for ATP" + sq.houseprice;
1615             }
1616             if (sq.hotel == 1) {
1617                 $("#buyhousebutton").hide();
1618                 sellhousebutton.value = "Sell hotel (ATP" + (sq.houseprice * 0.5) + ")";
1619                 sellhousebutton.title = "Sell a hotel for ATP" + (sq.houseprice * 0.5);
1620             }
1621
1622             var maxhouse = 0;
1623             var minhouse = 5;
1624
1625             for (var j = 0; j < max; j++) {
1626
1627                 if (square[currentSquare.group[j]].house > 0) {
1628                     allGroupUnimproved = false;
1629                     break;
1630                 }
1631             }
1632
1633             var max = sq.group.length;
1634             for (var i = 0; i < max; i++) {
1635                 s = square[sq.group[i]];
1636
1637                 if (s.owner !== sq.owner) {
1638                     buyhousebutton.disabled = true;
1639                     sellhousebutton.disabled = true;
1640                     buyhousebutton.title = "Before you can buy a house, you must own all
1641 the properties of this color-group.";
1642                 } else {
1643

```

```
1639         if (s.house > maxhouse) {
1640             maxhouse = s.house;
1641         }
1642
1643         if (s.house < minhouse) {
1644             minhouse = s.house;
1645         }
1646
1647         if (s.house > 0) {
1648             allGroupUnimproved = false;
1649         }
1650
1651         if (s.mortgage) {
1652             allGroupUnmortgaged = false;
1653         }
1654     }
1655 }
1656
1657 if (!allGroupUnmortgaged) {
1658     buyhousebutton.disabled = true;
1659     buyhousebutton.title = "Before you can buy a house, you must unmortgage
all the properties of this color-group.";
1660 }
1661
1662 // Force even building
1663 if (sq.house > minhouse) {
1664     buyhousebutton.disabled = true;
1665
1666     if (sq.house == 1) {
1667         buyhousebutton.title = "Before you can buy another house, the other
properties of this color-group must all have one house.";
1668     } else if (sq.house == 4) {
1669         buyhousebutton.title = "Before you can buy a hotel, the other
properties of this color-group must all have 4 houses.";
1670     } else {
1671         buyhousebutton.title = "Before you can buy a house, the other
properties of this color-group must all have " + sq.house + " houses.";
1672     }
1673 }
1674 if (sq.house < maxhouse) {
1675     sellhousebutton.disabled = true;
1676
1677     if (sq.house == 1) {
1678         sellhousebutton.title = "Before you can sell house, the other
properties of this color-group must all have one house.";
1679     } else {
1680         sellhousebutton.title = "Before you can sell a house, the other
properties of this color-group must all have " + sq.house + " houses.";
1681     }
1682 }
1683
1684 if (sq.house === 0 && sq.hotel === 0) {
1685     $("#sellhousebutton").hide();
1686
1687 } else {
1688     $("#mortgagebutton").hide();
1689 }
1690
1691
1692 // Before a property can be mortgaged or sold, all the properties of its
color-group must unimproved.
1693 if (!allGroupUnimproved) {
```

```

1694         document.getElementById("mortgagebutton").title = "Before a property can
be mortgaged, all the properties of its color-group must unimproved.";
1695         document.getElementById("mortgagebutton").disabled = true;
1696     }
1697
1698     } else {
1699         $("#buyhousebutton").hide();
1700         $("#sellhousebutton").hide();
1701     }
1702 }
1703 }
1704
1705 function landspecialstation() {
1706     var p = player[turn];
1707
1708     // Check if the player landed on a special station
1709     if (p.position === 2 || p.position === 4 || p.position === 6 || p.position === 23) {
1710         var cardIndex = specialCards.index; // Store current card index
1711         var drawnCard = specialCards[cardIndex]; // Get the drawn card
1712
1713         popup("<img src='images/special-station-card.png' style='height: 425px;
width: 250px; margin: 8px 8px 8px 8px; display: block; ' /> <div style='font-weight:
bold; font-size: 16px;'>Special Card:</div><div style='text-align: center; font-size:
16px; position: absolute;top: 40%; left: 48%; transform: translate(-50%, -50%); margin:
8px 8px 8px 8px; display: block;'>" + drawnCard.text + "</div>", function() {
1714             applySpecialCardEffect(cardIndex, p); // Apply the card effect
immediately
1715         }
1716     );
1717
1718     // Move to the next special card
1719     specialCards.index++;
1720
1721     // Reset index if we reach the end of the deck
1722     if (specialCards.index >= specialCards.length) {
1723         specialCards.index = 0;
1724     }
1725
1726     // Enzyme Cards
1727     } else if (p.position === 1 || p.position === 3 || p.position === 5
1728         || p.position === 7 || p.position === 8 || p.position === 9
1729         || p.position === 11 || p.position === 12 || p.position === 13 || p.position ===
14
1730         || p.position === 15 || p.position === 16 || p.position === 18 || p.position ===
19
1731         || p.position === 20 || p.position === 21 || p.position === 22 || p.position ===
24
1732         || p.position === 25 || p.position === 26 || p.position === 28 || p.position ===
29
1733         || p.position === 30 || p.position === 31 || p.position === 32 || p.position ===
33) {
1734         var enzymeCardsIndex = enzymeCards.deck[enzymeCards.index];
1735
1736         // Remove the get out of Enzyme Inhibition Zone free card from the deck.
1737         if (enzymeCardsIndex === 0) {
1738             enzymeCards.deck.splice(enzymeCards.index, 1);
1739         }
1740
1741         popup("<img src='images/enzyme-card.png' style='height: 425px; width: 250px;
margin: 8px 8px 8px 8px; display: block; ' /><div style='font-weight: bold; font-size:
16px;'>Cofactor Card:</div><div style='text-align: center; font-size: 16px; position:
absolute;top: 40%; left: 48%; transform: translate(-50%, -50%); margin: 8px 8px 8px 8px;
display: block;'>" + enzymeCards[enzymeCardsIndex].text + "</div>", function() {

```

```

1742         drawEnzymeCard(p, cardIndex);
1743     });
1744
1745     enzymeCards.index++;
1746
1747     if (enzymeCards.index >= enzymeCards.deck.length) {
1748         enzymeCards.index = 0;
1749     }
1750 } else {
1751     if (!p.human) {
1752         p.AI.alertList = "";
1753
1754         if (!p.AI.onLand()) {
1755             game.next();
1756         }
1757     }
1758 }
1759 }
1760
1761 function updateBoardUI() {
1762     for (let i = 0; i < square.length; i++) {
1763         let cell = document.getElementById("cell" + i + "name");
1764         if (cell) {
1765             cell.innerHTML = square[i].image ? `

```

```
1800     }
1801
1802     if (validStations.includes(p.position)) {
1803         applyCardEffect(cardIndex, player);
1804         alert(`${card.text} effect applied!`);
1805     } else {
1806         alert("Cofactor doesn't match the enzyme station.");
1807     }
1808 }
1809
1810 function applyCardEffect(cardIndex, player) {
1811     switch (cardIndex) {
1812
1813         case 0: // CoASH (Pyruvate dehydrogenase, Station 33 and a-ketoglutarate
dehydrogenase complex, Station 1)
1814             alert("CoASH effect applied! Skip the next rent.");
1815             player.skipNextRent = true;
1816             break;
1817
1818         case 1: // CoASH Negative Effect
1819             alert("CoASH negative effect! Lose one of your properties.");
1820             loseOneProperty(player);
1821             break;
1822
1823         case 2: // ADP (Pyruvate dehydrogenase, Station 33)
1824             alert("ADP effect applied! Move forward 2 spaces.");
1825             movePlayerForward(player, 2);
1826             break;
1827
1828         case 3: // ADP Negative Effect
1829             alert("ADP negative effect! Move back 2 spaces.");
1830             movePlayerBack(player, 2);
1831             break;
1832
1833         case 4: // Mg2+ (Hexokinase, Station 25 and Phosphoglycerate kinase, Station 23
and Pyruvate kinase, Station 24 and G6P, Station 11 and DNA polymerase, Stations 27-35,
and Pyruvate Carboxylase, Station 13)
1834             alert("Mg2+ effect applied! Move forward 1 space.");
1835             movePlayerForward(player, 1);
1836             break;
1837
1838         case 5: // Mg2+ Negative Effect
1839             alert("Mg2+ negative effect! Move back 1 space.");
1840             movePlayerBack(player, 1);
1841             break;
1842
1843         case 6: // Biotin (Pyruvate Carboxylase, Station 13)
1844             alert("Biotin effect applied! Roll the dice again.");
1845             rollDiceAgain(player, 1);
1846             break;
1847
1848         case 7: // Biotin Negative Effect
1849             alert("Biotin negative effect! Freeze for the next 2 turns.");
1850             player.freezeTurns = 2;
1851             break;
1852
1853         case 8: // GTP (PEPCK, Station 12)
1854             alert("GTP effect applied! Double your rent for 1 round.");
1855             player.doubleNextRent = true;
1856             break;
1857
```

```
1858     case 9: // GTP Negative Effect
1859         alert("GTP negative effect! Lose 20 ATP points.");
1860         player.atp -= 20;
1861         updateATP(player);
1862     break;
1863
1864     case 10: // CO2 (Pyruvate Carboxylase, Station 13)
1865         alert("CO2 effect applied! Skip the next rent.");
1866         player.skipNextRent = true;
1867     break;
1868
1869     case 11: // CO2 Negative Effect
1870         alert("CO2 negative effect! Move back 4 spaces.");
1871         movePlayerBack(player, 4);
1872     break;
1873
1874     case 12: // cAMP (Hormone-Sensitive Lipase (HSL), Station 9)
1875         alert("cAMP effect applied! Gain 20 ATP points.");
1876         player.atp += 20;
1877         updateATP(player);
1878     break;
1879
1880     case 13: // cAMP Negative Effect
1881         alert("cAMP negative effect! Go to Enzyme Inhibition Zone.");
1882         sendToInhibitionZone(player);
1883     break;
1884
1885     case 14: // FAD (a-ketoglutarate dehydrogenase complex, Station 1)
1886         alert("FAD effect applied! Move forward 4 spaces.");
1887         movePlayerForward(player, 4);
1888     break;
1889
1890     case 15: // FAD Negative Effect
1891         alert("FAD negative effect! Freeze for the next turn.");
1892         player.freezeTurns = 1;
1893     break;
1894
1895     case 16: // UDP Glucose (Glycogen synthase, Station 22)
1896         alert("UDP Glucose effect applied! Roll the dice again.");
1897         rollDiceAgain(player, 1);
1898     break;
1899
1900     case 17: // UDP Glucose Negative Effect
1901         alert("UDP Glucose negative effect! Go to Enzyme Inhibition Zone.");
1902         sendToInhibitionZone(player);
1903     break;
1904
1905     case 18: // Pi (Glycogen Phosphorylase, Station 23)
1906         alert("Pi effect applied! Draw an additional enzyme card.");
1907         drawEnzymeCard(player);
1908     break;
1909
1910     case 19: // Pi Negative Effect
1911         alert("Pi negative effect! Lose 5 ATP.");
1912         player.atp -= 5;
1913         updateATP(player);
1914     break;
1915
1916     case 20: // NAD+ (a-KG complex, Station 1)
1917         alert("NAD+ effect applied! Collect 10 ATP from each player.");
```

```
1918         collectFromEachPlayer(player, 10);
1919     break;
1920
1921     case 21: // NAD+ (Negative Effect)
1922         alert("NAD+ negative effect! Pay 15 ATP to the previous player.");
1923         payToPreviousPlayer(player, 15);
1924         break;
1925
1926     case 22: // PARP-1 (DNA polymerase beta, Station 28)
1927         alert("PARP-1 effect applied! Draw 2 cards on your next enzyme station.");
1928         player.drawTwoNextStation = true;
1929         break;
1930
1931     case 23: // PARP-1 (Negative Effect)
1932         alert("PARP-1 negative effect! Pay 15 ATP to the next player.");
1933         payToNextPlayer(player, 15);
1934         break;
1935
1936     case 24: // TPP (a-KG complex, Station 1)
1937         alert("TPP effect applied! Gain 10 ATP points.");
1938         player.atp += 10;
1939         updateATP(player);
1940         break;
1941
1942     case 25: // TPP (Negative Effect)
1943         alert("TPP negative effect! Can't buy your next enzyme station.");
1944
1945         break;
1946
1947     case 26: // NADPH (Billverdin reductase, Station 7)
1948         alert("NADPH effect applied! Collect 10 ATP from each player.");
1949         collectFromEachPlayer(player, 10);
1950         break;
1951
1952     case 27: // NADPH (Negative Effect)
1953         alert("NADPH negative effect! Lose 10 ATP points.");
1954         player.atp -= 10;
1955         updateATP(player);
1956         break;
1957
1958     case 28: // Transcription Factors (RNA Polymerase, Station 5)
1959         alert("Transcription Factors effect applied! Draw an additional enzyme card."
1960 );
1961         drawEnzymeCard(player);
1962         break;
1963
1964     case 29: // Transcription Factors (Negative Effect)
1965         alert("Transcription Factors negative effect! Freeze for the next 2 turns.");
1966         player.freezeTurns = 2;
1967         break;
1968
1969     case 30: // Ca2+ (Pyruvate Dehydrogenase & Glycogen Phosphorylase)
1970         alert("Ca2+ effect applied! Draw two cards on your next enzyme station.");
1971         player.drawTwoNextStation = true;
1972         break;
1973
1974     case 31: // Ca2+ Negative Effect
1975         alert("Ca2+ negative effect! You won't receive ATP next round.");
1976         player.skipNextATP = true;
1977         break;
```



```
1977
1978     case 32: // Lipoic Acid (a-KG complex, Station 1)
1979         alert("Lipoic Acid effect! Collect rent of your current station from each
player.");
1980         collectStationRentFromEachPlayer(player);
1981         break;
1982
1983     case 33: // Lipoic Acid Negative Effect
1984         alert("Lipoic Acid negative effect! Move back 1 space.");
1985         movePlayerBack(player, 1);
1986         break;
1987
1988     case 34: // SAM (Methyl Transferase, Station 16)
1989         alert("SAM effect! Roll the dice again twice.");
1990         rollDiceAgain(player, 2);
1991         break;
1992
1993     case 35: // SAM Negative Effect
1994         alert("SAM negative effect! Freeze for the next 3 turns.");
1995         player.freezeTurns = 3;
1996         break;
1997
1998     default:
1999         alert("Effect applied!");
2000 }
2001 }
2002
2003 function movePlayerBack(player, spaces) {
2004     player.position = Math.max(0, player.position - spaces);
2005     updatePosition(player);
2006     updateBoardUI();
2007 }
2008
2009 // Moves a player forward a number of spaces
2010 function movePlayerForward(player, spaces) {
2011     player.position = (player.position + spaces) % square.length;
2012     updatePosition(player);
2013     updateBoardUI();
2014 }
2015
2016 function collectFromEachPlayer(player, amount, pcount) {
2017     pcount.forEach(p => {
2018         if (p !== player) {
2019             p.atp -= amount;
2020             player.atp += amount;
2021         }
2022     });
2023     updateATP(player);
2024 }
2025
2026 // Pays ATP to the previous player
2027 function payToPreviousPlayer(player, amount) {
2028     let previousPlayer = pcount[(player[turn] - 1 + pcount.length) % pcount.length];
2029     player.atp -= amount;
2030     previousPlayer.atp += amount;
2031     updateATP(player);
2032     updateATP(previousPlayer);
2033 }
2034
2035 // Pays ATP to the next player
```



```
2036 function payToNextPlayer(player, amount) {
2037     let nextPlayer = pcount[(player[turn] + 1) % pcount.length];
2038     player.atp -= amount;
2039     nextPlayer.atp += amount;
2040     updateATP(player);
2041     updateATP(nextPlayer);
2042 }
2043
2044 // Moves the player to the Enzyme Inhibition Zone (Station 10)
2045 function sendToInhibitionZone(player) {
2046     alert(player.name + " has been sent to the Enzyme Inhibition Zone!");
2047     player.position = 10;
2048     updatePosition(player);
2049 }
2050
2051 // Player loses one property randomly
2052 function loseOneProperty(player) {
2053     if (player.properties.length > 0) {
2054         let lostProperty = player.properties.splice(Math.floor(Math.random() *
2055 player.properties.length), 1)[0];
2056         lostProperty.owner = null; // Unowned again
2057         alert(player.name + " lost " + lostProperty.name + "!");
2058     } else {
2059         alert("No properties to lose.");
2060     }
2061 }
2062
2063 // Player collects rent from all players for their current station
2064 function collectStationRentFromEachPlayer(player) {
2065     let station = square[player.position];
2066
2067     if (station.owner !== player) {
2068         alert("This station is not owned by you.");
2069         return;
2070     }
2071
2072     let rentAmount = station.rent[0]; // Base rent
2073     pcount.forEach(p => {
2074         if (p !== player) {
2075             p.atp -= rentAmount;
2076             player.atp += rentAmount;
2077         }
2078     });
2079     updateATP(player);
2080 }
2081
2082 // Rolls the dice again for a given number of times
2083 function rollDiceAgain(player, times, rollDice) {
2084     for (let i = 0; i < times; i++) {
2085         let roll = rollDice();
2086         player.position = (player.position + roll) % square.length;
2087     }
2088     updatePosition(player);
2089 }
2090
2091 function chanceAction(enzymeCardsIndex) {
2092     var p = player[turn]; // This is needed for reference in action() method.
2093
2094     // $('#popupbackground').hide();
2095     // $('#popupwrap').hide();
```

```
2095     enzymeCards[enzymeCardsIndex].action(p);
2096
2097     updateATP();
2098
2099     if (enzymeCardsIndex !== 35 && !p.human) {
2100         p.AI.alertList = "";
2101         game.next();
2102     }
2103 }
2104
2105 // Special Cards
2106 function applySpecialCardEffect(cardIndex, player) {
2107     switch (cardIndex) {
2108         case 0: // Insulin Stimulation
2109             alert("Insulin Stimulation! Gain 25 ATP and draw another special card.");
2110             player.atp += 25;
2111             updateATP(player);
2112             drawSpecialCard(p);
2113             break;
2114
2115         case 1: // Insulin Inhibition
2116             alert("Insulin Inhibition! Miss your next turn and lose 15 ATP.");
2117             player.missNextTurn = true;
2118             player.atp -= 15;
2119             updateATP(player);
2120             break;
2121
2122         case 2: // Glucagon Inhibition
2123             alert("Glucagon Inhibition! Reversing the effect of your last turn.");
2124             reverseLastEffect(player);
2125             break;
2126
2127         case 3: // Glucagon Stimulation
2128             alert("Glucagon Stimulation! Combine two enzyme stations and double ATP for 3
2129 turns.");
2130             player.combineStations = true;
2131             player.doublePathwayATP = 3;
2132             break;
2133
2134         case 4: // Increased Metabolic Rate
2135             alert("Increased Metabolic Rate! Double ATP rent for 3 turns.");
2136             player.doubleRentTurns = 3;
2137             break;
2138
2139         case 5: // Feedback Inhibition
2140             alert("Feedback Inhibition! You cannot draw enzyme cards for 2 turns.");
2141             player.noDrawTurns = 2;
2142             break;
2143
2144         case 6: // Epinephrine Stimulation
2145             alert("Epinephrine Stimulation! Instantly gain 10 ATP.");
2146             player.atp += 10;
2147             updateATP(player);
2148             break;
2149
2150         case 7: // Epinephrine Inhibition
2151             alert("Epinephrine Inhibition! Lose 15 ATP instantly.");
2152             player.atp -= 15;
2153             updateATP(player);
2154             break;
```

```
2154
2155     case 8: // Lactic Acidosis
2156         alert("Lactic Acidosis! You lose two enzyme stations.");
2157         loseTwoStations(player);
2158         break;
2159
2160     default:
2161         alert("Unknown special card effect.");
2162 }
2163 }
2164
2165 function drawSpecialCard(p) {
2166     var p = player[turn] // Get current player
2167     let cardIndex = Math.floor(Math.random() * specialCards.length); // Randomly pick a
card
2168     let card = specialCards[cardIndex];
2169
2170     alert("You drew a Special Card: " + card.name);
2171     applySpecialCardEffect(cardIndex, p);
2172 }
2173
2174 function reverseLastEffect(player) {
2175     alert(player.name + " is reversing the last turn's effect!");
2176     // Implement logic for reversing last effect
2177 }
2178
2179 function loseTwoStations(p) {
2180     if (player.properties.length < 2) {
2181         alert(player.name + " doesn't have enough stations to lose.");
2182         return;
2183     }
2184
2185     // Remove two enzyme stations randomly
2186     for (let i = 0; i < 2; i++) {
2187         let lostStation = player.properties.pop();
2188         lostStation.owner = null;
2189         alert(player.name + " lost " + lostStation.name);
2190     }
2191 }
2192
2193
2194
2195 function specialstationAction(specialstationIndex) {
2196     var p = player[turn]; // This is needed for reference in action() method.
2197
2198     $('#popupbackground').hide();
2199     $('#popupwrap').hide();
2200     specialCards[specialstationIndex].action(p);
2201
2202     updateATP();
2203
2204     if (specialstationIndex !== 8 && !p.human) {
2205         p.AI.alertList = "";
2206         game.next();
2207     }
2208 }
2209
2210 function addamount(amount, cause) {
2211     var p = player[turn];
2212 }
```

```
2213     p.money += amount;
2214
2215     addAlert(p.name + " received ATP" + amount + " from " + cause + ".");
2216 }
2217
2218 function subtractamount(amount, cause) {
2219     var p = player[turn];
2220
2221     p.pay(amount, 0);
2222
2223     addAlert(p.name + " lost ATP" + amount + " from " + cause + ".");
2224 }
2225
2226 function gotojail() {
2227     var p = player[turn];
2228     addAlert(p.name + " was sent directly to Enzyme Inhibition Zone.");
2229     document.getElementById("landed").innerHTML = "You are in Enzyme Inhibition Zone.";
2230
2231     p.jail = true;
2232     doublecount = 0;
2233
2234     document.getElementById("nextbutton").value = "End turn";
2235     document.getElementById("nextbutton").title = "End turn and advance to the next
2236 player.";
2237
2238     if (p.human) {
2239         document.getElementById("nextbutton").focus();
2240     }
2241
2242     updatePosition();
2243     updateOwned();
2244
2245     if (!p.human) {
2246         popup(p.AI.alertList, game.next);
2247         p.AI.alertList = "";
2248     }
2249 }
2250
2251 function gobackthreespaces() {
2252     var p = player[turn];
2253
2254     p.position -= 3;
2255
2256     land();
2257 }
2258
2259 function payeachplayer(amount, cause) {
2260     var p = player[turn];
2261     var total = 0;
2262
2263     for (var i = 1; i <= pcount; i++) {
2264         if (i !== turn) {
2265             player[i].money += amount;
2266             total += amount;
2267             creditor = p.money >= 0 ? i : creditor;
2268
2269             p.pay(amount, creditor);
2270         }
2271     }
```

```
2272     addAlert(p.name + " lost ATP" + total + " from " + cause + ".");
2273 }
2274
2275 function collectfromeachplayer(amount, cause) {
2276     var p = player[turn];
2277     var total = 0;
2278
2279     for (var i = 1; i <= pcount; i++) {
2280         if (i != turn) {
2281             money = player[i].money;
2282             if (money < amount) {
2283                 p.money += money;
2284                 total += money;
2285                 player[i].money = 0;
2286             } else {
2287                 player[i].pay(amount, turn);
2288                 p.money += amount;
2289                 total += amount;
2290             }
2291         }
2292     }
2293
2294     addAlert(p.name + " received ATP" + total + " from " + cause + ".");
2295 }
2296
2297 function advance(destination, pass) {
2298     var p = player[turn];
2299
2300     if (typeof pass === "number") {
2301         if (p.position < pass) {
2302             p.position = pass;
2303         } else {
2304             p.position = pass;
2305             p.money += 200;
2306             addAlert(p.name + " collected a $200 salary for passing GO.");
2307         }
2308     }
2309     if (p.position < destination) {
2310         p.position = destination;
2311     } else {
2312         p.position = destination;
2313         p.money += 200;
2314         addAlert(p.name + " collected a $200 salary for passing GO.");
2315     }
2316
2317     land();
2318 }
2319
2320 function advanceToNearestUtility() {
2321     var p = player[turn];
2322
2323     if (p.position < 12) {
2324         p.position = 12;
2325     } else if (p.position >= 12 && p.position < 28) {
2326         p.position = 28;
2327     } else if (p.position >= 28) {
2328         p.position = 12;
2329         p.money += 200;
2330         addAlert(p.name + " collected a $200 salary for passing GO.");
2331     }
```

```
2332
2333     land(true);
2334 }
2335
2336 function advanceToNearestRailroad() {
2337     var p = player[turn];
2338
2339     updatePosition();
2340
2341     if (p.position < 15) {
2342         p.position = 15;
2343     } else if (p.position >= 15 && p.position < 25) {
2344         p.position = 25;
2345     } else if (p.position >= 35) {
2346         p.position = 5;
2347         p.money += 200;
2348         addAlert(p.name + " collected a $200 salary for passing GO.");
2349     }
2350
2351     land(true);
2352 }
2353
2354 function streetrepairs(houseprice, hotelprice) {
2355     var cost = 0;
2356     for (var i = 0; i < 34; i++) {
2357         var s = square[i];
2358         if (s.owner == turn) {
2359             if (s.hotel == 1)
2360                 cost += hotelprice;
2361             else
2362                 cost += s.house * houseprice;
2363         }
2364     }
2365
2366     var p = player[turn];
2367
2368     if (cost > 0) {
2369         p.pay(cost, 0);
2370
2371         // If function was called by Community Chest.
2372         if (houseprice === 40) {
2373             addAlert(p.name + " lost ATP" + cost + " to Community Chest.");
2374         } else {
2375             addAlert(p.name + " lost ATP" + cost + " to Chance.");
2376         }
2377     }
2378 }
2379
2380
2381 function payfifty() {
2382     var p = player[turn];
2383
2384     document.getElementById("jail").style.border = '1px solid black';
2385     document.getElementById("cell11").style.border = '2px solid ' + p.color;
2386
2387     $("#landed").hide();
2388     doublecount = 0;
2389
2390     p.jail = false;
2391     p.jailroll = 0;
```

```
2392     p.position = 10;
2393     p.pay(50, 0);
2394
2395     addAlert(p.name + " paid the 50 ATP fine to get out of enzyme inhibition zone.");
2396     updateATP();
2397     updatePosition();
2398 }
2399
2400 function useJailCard() {
2401     var p = player[turn];
2402
2403     document.getElementById("jail").style.border = '1px solid black';
2404     document.getElementById("cell11").style.border = '2px solid ' + p.color;
2405
2406     $("#landed").hide();
2407     p.jail = false;
2408     p.jailroll = 0;
2409
2410     p.position = 10;
2411
2412     doublecount = 0;
2413
2414     if (p.specialstationJailCard) {
2415         p.specialstationJailCard = false;
2416
2417         // Insert the get out of jail free card back into the community chest deck.
2418         specialCards.deck.splice(specialCards.index, 0, 0);
2419
2420         specialCards.index++;
2421
2422         if (specialCards.index >= specialCards.deck.length) {
2423             specialCards.index = 0;
2424         }
2425     } else if (p.chanceJailCard) {
2426         p.chanceJailCard = false;
2427
2428         // Insert the get out of jail free card back into the chance deck.
2429         enzymeCards.deck.splice(enzymeCards.index, 0, 0);
2430
2431         enzymeCards.index++;
2432
2433         if (enzymeCards.index >= enzymeCards.deck.length) {
2434             enzymeCards.index = 0;
2435         }
2436     }
2437
2438     addAlert(p.name + " used a \"Get Out of Enzyme Inhibition Zone Free\" card.");
2439     updateOwned();
2440     updatePosition();
2441 }
2442
2443 function buyHouse(index) {
2444     var sq = square[index];
2445     var p = player[sq.owner];
2446     var houseSum = 0;
2447     var hotelSum = 0;
2448
2449     if (p.money - sq.houseprice < 0) {
2450         if (sq.house == 4) {
2451             return false;
```

```
2452         } else {
2453             return false;
2454         }
2455
2456     } else {
2457         for (var i = 0; i < 34; i++) {
2458             if (square[i].hotel === 1) {
2459                 hotelSum++;
2460             } else {
2461                 houseSum += square[i].house;
2462             }
2463         }
2464
2465         if (sq.house < 4) {
2466             if (houseSum >= 32) {
2467                 return false;
2468
2469             } else {
2470                 sq.house++;
2471                 addAlert(p.name + " placed a house on " + sq.name + ".");
2472             }
2473
2474         } else {
2475             if (hotelSum >= 12) {
2476                 return;
2477
2478             } else {
2479                 sq.house = 5;
2480                 sq.hotel = 1;
2481                 addAlert(p.name + " placed a hotel on " + sq.name + ".");
2482             }
2483         }
2484
2485         p.pay(sq.houseprice, 0);
2486
2487         updateOwned();
2488         updateATP();
2489     }
2490 }
2491
2492 function sellHouse(index) {
2493     sq = square[index];
2494     p = player[sq.owner];
2495
2496     if (sq.hotel === 1) {
2497         sq.hotel = 0;
2498         sq.house = 4;
2499         addAlert(p.name + " sold the hotel on " + sq.name + ".");
2500     } else {
2501         sq.house--;
2502         addAlert(p.name + " sold a house on " + sq.name + ".");
2503     }
2504
2505     p.money += sq.houseprice * 0.5;
2506     updateOwned();
2507     updateATP();
2508 }
2509
2510 function showStats() {
2511     console.log("✅ showStats() function started.");
```



```

2512
2513     var HTML, sq, p;
2514     var mortgagetext, housetext;
2515     var write;
2516     HTML = "<table align='center'><tr>";
2517
2518     for (var x = 1; x <= pcount; x++) {
2519         if (!player[x]) {
2520             console.error(`❌ Player ${x} not found!`);
2521             continue;
2522         }
2523
2524         write = false;
2525         p = player[x];
2526         console.log("💠 Processing player:", p.name);
2527
2528         HTML += `<td class='statscell' id='statscell${x}' style='border: 2px solid
2529     ${p.color}'>
2530             <div class='statsplayername'>${p.name}</div>`;
2531
2532         for (var i = 0; i < 34; i++) {
2533             sq = square[i];
2534
2535             if (sq.owner === x) {
2536                 console.log("🏠 Processing property:", sq.name);
2537
2538                 mortgagetext = sq.mortgage ? "title='Mortgaged' style='color: grey;'" : "
2539 ";
2540                 housetext = "";
2541
2542                 if (!write) {
2543                     write = true;
2544                     HTML += "<table>";
2545                 }
2546
2547                 HTML += `<tr>
2548                     <td class='statscellcolor' style='background: ${sq.color};'
2549     onmouseover='showdeed(${i});' onmouseout='hidedeed();'></td>
2550                     <td class='statscellname' ${mortgagetext}>${sq.name}
2551     ${housetext}</td>
2552                     </tr>`;
2553             }
2554         }
2555
2556         if (!write) {
2557             HTML += ` ${p.name} doesn't have any properties.`;
2558         } else {
2559             HTML += "</table>";
2560         }
2561
2562         HTML += "</td>";
2563     }
2564
2565     HTML += "</tr></table><div id='titled deed'></div>";
2566
2567     console.log("🚩 Final HTML:", HTML);
2568     document.getElementById("statstext").innerHTML = HTML;
2569
2570     console.log("✅ Stats updated in statstext div.");

```

```

2569
2570 // Show using animation.
2571 $("#statsbackground").fadeIn(400, function() {
2572     console.log("📊 Stats popup displayed.");
2573     $("#statswrap").show();
2574 });
2575 }
2576
2577
2578
2579 function showdeed(property) {
2580     var sq = square[property];
2581     $("#deed").show();
2582
2583     $("#deed-normal").hide();
2584     $("#deed-mortgaged").hide();
2585     $("#deed-special").hide();
2586
2587     console.log("Hovered Property Index:", property);
2588     console.log("Square Data:", square[property]);
2589
2590     var deedElement = document.getElementById("deed");
2591
2592     if (!deedElement) return;
2593
2594     deedElement.style.display = "block";
2595     deedElement.style.top = event.clientY + 10 + "px";
2596     deedElement.style.left = event.clientX + 10 + "px";
2597
2598     if (!square[property]) {
2599         console.error("Error: Property data missing for index", property);
2600         return;
2601     }
2602     console.log("📍 Station Data:", square[property]);
2603
2604     if (sq.mortgage) {
2605         $("#deed-mortgaged").show();
2606         document.getElementById("deed-mortgaged-name").textContent = sq.name;
2607         document.getElementById("deed-mortgaged-mortgage").textContent = (sq.price / 2);
2608
2609     } else {
2610
2611         if (sq.groupNumber >= 3) {
2612             $("#deed-normal").show();
2613             document.getElementById("deed-header").style.backgroundColor = sq.color;
2614             document.getElementById("deed-name").textContent = sq.name;
2615             document.getElementById("deed-price").textContent = "Purchase ATP " +
sq.price;
2616             document.getElementById("deed-baserent").textContent = "Rent ATP " +
sq.baserent;
2617             document.getElementById("deed-rent1").textContent = sq.rent1;
2618             document.getElementById("deed-rent2").textContent = sq.rent2;
2619             document.getElementById("deed-rent3").textContent = sq.rent3;
2620             document.getElementById("deed-rent4").textContent = sq.rent4;
2621             document.getElementById("deed-rent5").textContent = sq.rent5;
2622             document.getElementById("deed-mortgage").textContent = (sq.price / 2);
2623         } else if (sq.groupNumber == 2) {
2624             $("#deed-special").show();
2625             document.getElementById("deed-special").style.backgroundColor = sq.color;
2626             document.getElementById("deed-special-name").textContent = sq.name;

```

```
2627     document.getElementById("deed-special-text").textContent = sq.pricetext;
2628     document.getElementById("deed-special-footer").textContent = "";
2629     // Clear price & rent for special and corner stations
2630     document.getElementById("deed-price").textContent = "";
2631     document.getElementById("deed-baserent").textContent = "";
2632
2633     } else if (sq.groupNumber >= 9) {
2634         console.log("🔍 Handling groupNumber >= 9");
2635
2636         $("#deed-normal").show();
2637         document.getElementById("deed-header").style.backgroundColor = sq.color
|| "gray";
2638         document.getElementById("deed-name").textContent = sq.name;
2639
2640         // Use pricetext instead of price & baserent
2641         document.getElementById("deed-price").textContent = sq.pricetext || "";
2642         document.getElementById("deed-baserent").textContent = "";
2643
2644     } else if (sq.groupNumber === 0) {
2645         $("#deed-normal").show();
2646         document.getElementById("deed-header").style.backgroundColor = sq.color;
2647         document.getElementById("deed-name").textContent = sq.name;
2648         document.getElementById("deed-price").textContent = sq.pricetext;
2649     }
2650 }
2651 }
2652
2653 function hidedeed() {
2654     $("#deed").hide();
2655 }
2656
2657 function buy() {
2658     var p = player[turn];
2659     var property = square[p.position];
2660     var cost = property.price;
2661
2662     if (p.money >= cost) {
2663         p.pay(cost, 0);
2664
2665         property.owner = turn;
2666         updateATP();
2667         addAlert(p.name + " bought " + property.name + " for " + property.pricetext + ".");
2668     };
2669
2670     updateOwned();
2671
2672     $("#landed").hide();
2673
2674     } else {
2675         popup("<p>" + p.name + ", you need ATP" + (property.price - p.money) + " more to
buy " + property.name + "</p>");
2676     }
2677 }
2678
2679 function mortgage(index) {
2680     var sq = square[index];
2681     var p = player[sq.owner];
2682
2683     if (sq.house > 0 || sq.hotel > 0 || sq.mortgage) {
2684         return false;
2685     }
2686 }
```

```

2685
2686     var mortgagePrice = Math.round(sq.price * 0.5);
2687     var unmortgagePrice = Math.round(sq.price * 0.55);
2688
2689     sq.mortgage = true;
2690     p.money += mortgagePrice;
2691
2692     document.getElementById("mortgagebutton").value = "Unmortgage for ATP" +
unmortgagePrice;
2693     document.getElementById("mortgagebutton").title = "Unmortgage " + sq.name + " for
ATP" + unmortgagePrice + ".";
2694
2695     addAlert(p.name + " mortgaged " + sq.name + " for ATP" + mortgagePrice + ".");
2696     updateOwned();
2697     updateATP();
2698
2699     return true;
2700 }
2701
2702 function unmortgage(index) {
2703     var sq = square[index];
2704     var p = player[sq.owner];
2705     var unmortgagePrice = Math.round(sq.price * 0.55);
2706     var mortgagePrice = Math.round(sq.price * 0.5);
2707
2708     if (unmortgagePrice > p.money || !sq.mortgage) {
2709         return false;
2710     }
2711
2712     p.pay(unmortgagePrice, 0);
2713     sq.mortgage = false;
2714     document.getElementById("mortgagebutton").value = "Mortgage for ATP" + mortgagePrice;
2715     document.getElementById("mortgagebutton").title = "Mortgage " + sq.name + " for ATP"
+ mortgagePrice + ".";
2716
2717     addAlert(p.name + " unmortgaged " + sq.name + " for ATP" + unmortgagePrice + ".");
2718     updateOwned();
2719     return true;
2720 }
2721
2722
2723 function land(increasedRent) {
2724     increasedRent = !!increasedRent; // Cast increasedRent to a boolean value. It is used
for the ADVANCE TO THE NEAREST RAILROAD/UTILITY Chance cards.
2725
2726     var p = player[turn];
2727     var s = square[p.position];
2728
2729     var die1 = game.getDie(1);
2730     var die2 = game.getDie(2);
2731
2732     $("#landed").show();
2733     document.getElementById("landed").innerHTML = "You landed on " + s.name + ".";
2734     s.landcount++;
2735     addAlert(p.name + " landed on " + s.name + ".");
2736
2737     // Allow player to buy the property on which he landed.
2738     if (s.price !== 0 && s.owner === 0) {
2739
2740         if (!p.human) {
2741

```

```

2742         if (p.AI.buyProperty(p.position)) {
2743             buy();
2744         }
2745     } else {
2746         document.getElementById("landed").innerHTML = "<div>You landed on <a href='
javascript:void(0);' onmouseover='showdeed(" + p.position + ");' onmouseout='hidedeed()';'
class='statscellcolor'>" + s.name + "</a>.<input type='button' onclick='buy();' value='
Buy ($" + s.price + ")' title='Buy " + s.name + " for " + s.pricetext + ".'/></div>";
2747     }
2748
2749     game.addPropertyToAuctionQueue(p.position);
2750 }
2751
2752 // Collect rent
2753 if (s.owner !== 0 && s.owner !== turn && !s.mortgage) {
2754     var groupowned = true;
2755     var rent;
2756
2757     if (p.position === 12) {
2758         if (increasedRent || square[28].owner == s.owner) {
2759             rent = (die1 + die2) * 10;
2760         } else {
2761             rent = (die1 + die2) * 4;
2762         }
2763
2764     } else if (p.position === 28) {
2765         if (increasedRent || square[12].owner == s.owner) {
2766             rent = (die1 + die2) * 10;
2767         } else {
2768             rent = (die1 + die2) * 4;
2769         }
2770
2771     } else {
2772
2773         for (var i = 0; i < 34; i++) {
2774             sq = square[i];
2775             if (sq.groupNumber == s.groupNumber && sq.owner !== s.owner) {
2776                 groupowned = false;
2777             }
2778         }
2779
2780         if (!groupowned) {
2781             rent = s.baserent;
2782         } else {
2783             if (s.house === 0) {
2784                 rent = s.baserent * 2;
2785             } else {
2786                 rent = s["rent" + s.house];
2787             }
2788         }
2789     }
2790
2791     addAlert(p.name + " paid ATP" + rent + " rent to " + player[s.owner].name + ".");
2792     p.pay(rent, s.owner);
2793     player[s.owner].money += rent;
2794
2795     document.getElementById("landed").innerHTML = "You landed on " + s.name + ". " +
player[s.owner].name + " collected ATP" + rent + " rent.";
2796 } else if (s.owner > 0 && s.owner !== turn && s.mortgage) {
2797     document.getElementById("landed").innerHTML = "You landed on " + s.name + ".
Property is mortgaged; no rent was collected.";

```

```
2798     }
2799
2800
2801
2802     // Go to Enzyme Inhibition Zone. Go directly to Enzyme Inhibition Zone. Do not pass
GO. Do not collect $200.
2803     if (p.position === 30) {
2804         updateATP();
2805         updatePosition();
2806
2807         if (p.human) {
2808             popup("<div>Go to Enzyme Inhibition Zone. Go directly to Enzyme Inhibition
Zone. Do not pass GO. Do not collect $200.</div>", gotojail);
2809         } else {
2810             gotojail();
2811         }
2812
2813         return;
2814     }
2815
2816     // Luxury Tax
2817     if (p.position === 35) {
2818         luxurymtax();
2819     }
2820
2821     updateATP();
2822     updatePosition();
2823     updateOwned();
2824
2825     if (!p.human) {
2826         popup(p.AI.alertList, landspecialstation);
2827         p.AI.alertList = "";
2828     } else {
2829         landspecialstation();
2830     }
2831 }
2832
2833 function roll() {
2834     var p = player[turn];
2835
2836     $("#option").hide();
2837     $("#buy").show();
2838     $("#manage").hide();
2839
2840     if (p.human) {
2841         document.getElementById("nextbutton").focus();
2842     }
2843     document.getElementById("nextbutton").value = "End turn";
2844     document.getElementById("nextbutton").title = "End turn and advance to the next
player.";
2845
2846     game.rollDice();
2847     var die1 = game.getDie(1);
2848     var die2 = game.getDie(2);
2849
2850     doublecount++;
2851
2852     if (die1 == die2) {
2853         addAlert(p.name + " rolled " + (die1 + die2) + " - doubles.");
2854     } else {
2855         addAlert(p.name + " rolled " + (die1 + die2) + ".");
```

```
2856     }
2857
2858     if (die1 == die2 && !p.jail) {
2859         updateDice(die1, die2);
2860
2861         if (doublecount < 3) {
2862             document.getElementById("nextbutton").value = "Roll again";
2863             document.getElementById("nextbutton").title = "You threw doubles. Roll
again.";
2864
2865             // If player rolls doubles three times in a row, send him to Enzyme Inhibition
Zone
2866             } else if (doublecount === 3) {
2867                 p.jail = true;
2868                 doublecount = 0;
2869                 addAlert(p.name + " rolled doubles three times in a row.");
2870                 updateATP();
2871
2872                 if (p.human) {
2873                     popup("You rolled doubles three times in a row. Go to Enzyme Inhibition
Zone.", gotojail);
2874                 } else {
2875                     gotojail();
2876                 }
2877             }
2878
2879             return;
2880         }
2881     } else {
2882         document.getElementById("nextbutton").value = "End turn";
2883         document.getElementById("nextbutton").title = "End turn and advance to the next
player.";
2884         doublecount = 0;
2885     }
2886
2887     updatePosition();
2888     updateATP();
2889     updateOwned();
2890
2891     if (p.jail === true) {
2892         p.jailroll++;
2893
2894         updateDice(die1, die2);
2895         if (die1 == die2) {
2896             document.getElementById("jail").style.border = "1px solid black";
2897             document.getElementById("cell11").style.border = "2px solid " + p.color;
2898             $("#landed").hide();
2899
2900             p.jail = false;
2901             p.jailroll = 0;
2902             p.position = 10 + die1 + die2;
2903             doublecount = 0;
2904
2905             addAlert(p.name + " rolled doubles to get out of Enzyme Inhibition Zone.");
2906
2907             land();
2908         } else {
2909             if (p.jailroll === 3) {
2910
2911                 if (p.human) {
2912                     popup("<p>You must pay the $50 fine.</p>", function() {
```

```
2913         payfifty();
2914         player[turn].position=10 + die1 + die2;
2915         land();
2916     });
2917     } else {
2918         payfifty();
2919         p.position = 10 + die1 + die2;
2920         land();
2921     }
2922     } else {
2923         $("#landed").show();
2924         document.getElementById("landed").innerHTML = "You are in Enzyme
Inhibition Zone.";
2925
2926         if (!p.human) {
2927             popup(p.AI.alertList, game.next);
2928             p.AI.alertList = "";
2929         }
2930     }
2931 }
2932
2933
2934 } else {
2935     updateDice(die1, die2);
2936
2937     // Move player
2938     p.position += die1 + die2;
2939
2940     // Collect $200 salary as you pass GO
2941     if (p.position >= 40) {
2942         p.position -= 40;
2943         p.money += 200;
2944         addAlert(p.name + " collected a $200 salary for passing GO.");
2945     }
2946
2947     land();
2948 }
2949 }
2950
2951 function play() {
2952     if (game.auction()) {
2953         return;
2954     }
2955
2956     turn++;
2957     if (turn > pcount) {
2958         turn -= pcount;
2959     }
2960
2961     var p = player[turn];
2962     game.resetDice();
2963
2964     document.getElementById("pname").innerHTML = p.name;
2965
2966     addAlert("It is " + p.name + "'s turn.");
2967
2968     // Check for bankruptcy.
2969     p.pay(0, p.creditor);
2970
2971     $("#landed, #option, #manage").hide();
```



```

2972     $("#board, #control, #moneybar, #viewstats, #buy").show();
2973
2974     doublecount = 0;
2975     if (p.human) {
2976         document.getElementById("nextbutton").focus();
2977     }
2978     document.getElementById("nextbutton").value = "Roll Dice";
2979     document.getElementById("nextbutton").title = "Roll the dice and move your token
accordingly.";
2980
2981     $("#die0").hide();
2982     $("#die1").hide();
2983
2984     if (p.jail) {
2985         $("#landed").show();
2986         document.getElementById("landed").innerHTML = "You are in Enzyme Inhibition Zone.


```

```
3027     if (!p.human) {
3028         if (!p.AI.beforeTurn()) {
3029             game.next();
3030         }
3031     }
3032 }
3033
3034 function endGame() {
3035     let winner = player.reduce((prev, curr) => (curr.money > prev.money ? curr : prev),
3036 player[0]);
3037     alert(`Game Over! ${winner.name} wins with ATP ${winner.money}!`);
3038     document.getElementById("control").style.display = "none";
3039 }
3040
3041 function setup() {
3042     var roundLimit = parseInt(document.getElementById("roundLimitSelect").value, 10);
3043     game.roundLimit = roundLimit;
3044     game.currentRound = 0;
3045
3046
3047     pcount = parseInt(document.getElementById("playernumber").value, 10);
3048
3049     var playerArray = new Array(pcount);
3050     var p;
3051
3052     playerArray.randomize();
3053
3054     for (var i = 1; i <= pcount; i++) {
3055         p = player[playerArray[i - 1]];
3056
3057
3058         p.color = document.getElementById("player" + i + "color").value.toLowerCase();
3059
3060         if (document.getElementById("player" + i + "ai").value === "0") {
3061             p.name = document.getElementById("player" + i + "name").value;
3062             p.human = true;
3063         } else if (document.getElementById("player" + i + "ai").value === "1") {
3064             p.human = false;
3065             p.AI = new AITest(p);
3066         }
3067     }
3068
3069     $("#board, #moneybar").show();
3070     $("#setup").hide();
3071
3072     if (pcount === 2) {
3073         document.getElementById("stats").style.width = "454px";
3074     } else if (pcount === 3) {
3075         document.getElementById("stats").style.width = "686px";
3076     }
3077
3078     document.getElementById("stats").style.top = "0px";
3079     document.getElementById("stats").style.left = "0px";
3080     play();
3081     updateBoardUI();
3082 }
3083
3084 // function togglecheck(elementid) {
3085     // element = document.getElementById(elementid);
```

```
3086
3087 // if (window.event.srcElement.id == elementid)
3088 // return;
3089
3090 // if (element.checked) {
3091 // element.checked = false;
3092 // } else {
3093 // element.checked = true;
3094 // }
3095 // }
3096
3097 function getCheckedProperty() {
3098     for (var i = 0; i < 42; i++) {
3099         if (document.getElementById("propertycheckbox" + i) && document.getElementById("
propertycheckbox" + i).checked) {
3100             return i;
3101         }
3102     }
3103     return -1; // No property is checked.
3104 }
3105
3106 // function propertycell_onclick(element, num) {
3107 // togglecheck("propertycheckbox" + num);
3108 // if (document.getElementById("propertycheckbox" + num).checked) {
3109
3110 // // Uncheck all other boxes.
3111 // for (var i = 0; i < 34; i++) {
3112 //     if (i !== num && document.getElementById("propertycheckbox" + i)) {
3113 //         document.getElementById("propertycheckbox" + i).checked = false;
3114 //     }
3115 // }
3116 // }
3117
3118 // updateOption();
3119 // }
3120
3121 function playernumber_onchange() {
3122     pcount = parseInt(document.getElementById("playernumber").value, 10);
3123
3124     $(".player-input").hide();
3125
3126     for (var i = 1; i <= pcount; i++) {
3127         $("#player" + i + "input").show();
3128     }
3129 }
3130
3131 function menuitem_onmouseover(element) {
3132     element.className = "menuitem menuitem_hover";
3133     return;
3134 }
3135
3136 function menuitem_onmouseout(element) {
3137     element.className = "menuitem";
3138     return;
3139 }
3140
3141 window.onload = function() {
3142     game = new Game();
3143     updateBoardUI();
3144     for (var i = 0; i <= 8; i++) {
```

```
3145     player[i] = new Player("", "");
3146     player[i].index = i;
3147 }
3148
3149 var groupPropertyArray = [];
3150 var groupNumber;
3151
3152 for (var i = 0; i < 34; i++) {
3153     groupNumber = square[i].groupNumber;
3154
3155     if (groupNumber > 0) {
3156         if (!groupPropertyArray[groupNumber]) {
3157             groupPropertyArray[groupNumber] = [];
3158         }
3159
3160         groupPropertyArray[groupNumber].push(i);
3161     }
3162 }
3163
3164 for (var i = 0; i < 34; i++) {
3165     groupNumber = square[i].groupNumber;
3166
3167     if (groupNumber > 0) {
3168         square[i].group = groupPropertyArray[groupNumber];
3169     }
3170
3171     square[i].index = i;
3172 }
3173
3174 AITest.count = 0;
3175
3176 player[1].human = true;
3177 player[0].name = "the bank";
3178
3179 specialCards.index = 0;
3180 enzymeCards.index = 0;
3181 diseaseCards.index = 0;
3182
3183 specialCards.deck = [];
3184 enzymeCards.deck = [];
3185 diseaseCards.deck = [];
3186
3187 for (var i = 0; i < 9; i++) {
3188     specialCards.deck[i] = i;
3189 }
3190
3191 for (var i = 0; i < 36; i++) {
3192     enzymeCards.deck[i] = i;
3193 }
3194
3195 for (var i = 0; i < 20; i++) {
3196     diseaseCards.deck[i] = i;
3197 }
3198
3199 // Shuffle enzyme, special and disease decks.
3200 enzymeCards.deck.sort(function() {return Math.random() - 0.5;});
3201 specialCards.deck.sort(function() {return Math.random() - 0.5;});
3202 diseaseCards.deck.sort(function() {return Math.random() - 0.5;});
3203
3204 $("#playernumber").on("change", playernumber_onchange);
```

```
3205     playernumber_onchange();
3206
3207     $("#nextbutton").click(game.next);
3208     $("#noscript").hide();
3209     $("#setup, #noF5").show();
3210
3211     var enlargeWrap = document.body.appendChild(document.createElement("div"));
3212
3213     enlargeWrap.id = "enlarge-wrap";
3214
3215     var HTML = "";
3216     for (var i = 0; i < 34; i++) {
3217         HTML += "<div id='enlarge' + i + '' class='enlarge'>";
3218         HTML += "<div id='enlarge' + i + 'color' class='enlarge-color'></div><br /><div";
3219         HTML += "id='enlarge' + i + 'name' class='enlarge-name'></div>";
3220         HTML += "<br /><div id='enlarge' + i + 'price' class='enlarge-price'></div>";
3221         HTML += "<br /><div id='enlarge' + i + 'token' class='enlarge-token'></div></div>";
3222     }
3223
3224     enlargeWrap.innerHTML = HTML;
3225
3226     var currentCell;
3227     var currentCellAnchor;
3228     var currentCellPositionHolder;
3229     var currentCellName;
3230     var currentCellOwner;
3231
3232     for (var i = 0; i < 34; i++) {
3233         s = square[i];
3234
3235         currentCell = document.getElementById("cell" + i);
3236
3237         currentCellAnchor = currentCell.appendChild(document.createElement("div"));
3238         currentCellAnchor.id = "cell" + i + "anchor";
3239         currentCellAnchor.className = "cell-anchor";
3240
3241         currentCellPositionHolder =
3242         currentCellAnchor.appendChild(document.createElement("div"));
3243         currentCellPositionHolder.id = "cell" + i + "positionholder";
3244         currentCellPositionHolder.className = "cell-position-holder";
3245         currentCellPositionHolder.enlargeId = "enlarge" + i;
3246
3247         if (square[i].groupNumber) {
3248             currentCellOwner = currentCellAnchor.appendChild(document.createElement("div"
3249             ));
3250             currentCellOwner.id = "cell" + i + "owner";
3251             currentCellOwner.className = "cell-owner";
3252         }
3253
3254         document.getElementById("enlarge" + i + "color").style.backgroundColor = s.color;
3255         document.getElementById("enlarge" + i + "name").textContent = s.name;
3256         document.getElementById("enlarge" + i + "price").textContent = s.pricetext;
3257     }
3258
3259     // Add images to enlarges.
3260     document.getElementById("enlarge0token").innerHTML += '';
```

```

3261     document.getElementById("enlarge20price").innerHTML += "<img src='images/free-
oxidation-station.png' height='80' width='72' alt='' style='position: relative; top:
-20px;' />";
3262     document.getElementById("enlarge38token").innerHTML += '';
3263
3264     corrections();
3265
3266     // Jail corrections
3267     $("<div>", {id: "jailpositionholder" }).appendTo("#jail");
3268     $("<span>").text("Jail").appendTo("#jail");
3269
3270     document.getElementById("jail").enlargeId = "enlarge40";
3271
3272     document.getElementById("enlarge-wrap").innerHTML += "<div id='enlarge40' class='
enlarge'><div id='enlarge40color' class='enlarge-color'></div><br /><div id='
enlarge40name' class='enlarge-name'>Enzyme Inhibition Zone</div><br /><div id='
enlarge40price' class='enlarge-price'><img src='images/inhibition-zone.png' height='80'
width='80' alt='' style='position: relative; top: -20px;' /></div><br /><div id='
enlarge40token' class='enlarge-token'></div></div>";
3273
3274     document.getElementById("enlarge40name").innerHTML = "Enzyme Inhibition Zone";
3275
3276     // Create event handlers for hovering and dragging.
3277
3278     var drag, dragX, dragY, dragObj, dragTop, dragLeft;
3279
3280     $(".cell-position-holder, #jail").on("mouseover", function(){
3281         $("#" + this.enlargeId).show();
3282
3283     }).on("mouseout", function() {
3284         $("#" + this.enlargeId).hide();
3285
3286     }).on("mousemove", function(e) {
3287         var element = document.getElementById(this.enlargeId);
3288
3289         if (e.clientY + 20 > window.innerHeight - 204) {
3290             element.style.top = (window.innerHeight - 204) + "px";
3291         } else {
3292             element.style.top = (e.clientY + 20) + "px";
3293         }
3294
3295         element.style.left = (e.clientX + 10) + "px";
3296     });
3297
3298
3299     $("body").on("mousemove", function(e) {
3300         var object;
3301
3302         if (e.target) {
3303             object = e.target;
3304         } else if (window.event && window.event.srcElement) {
3305             object = window.event.srcElement;
3306         }
3307
3308
3309         if (object.classList.contains("propertycellcolor") || object.classList.contains("
statscellcolor")) {
3310             if (e.clientY + 20 > window.innerHeight - 279) {
3311                 document.getElementById("deed").style.top = (window.innerHeight - 279) +
"px";
3312             } else {

```

```
3313         document.getElementById("deed").style.top = (e.clientY + 20) + "px";
3314     }
3315     document.getElementById("deed").style.left = (e.clientX + 10) + "px";
3316
3317
3318 } else if (drag) {
3319     if (e) {
3320         dragObj.style.left = (dragLeft + e.clientX - dragX) + "px";
3321         dragObj.style.top = (dragTop + e.clientY - dragY) + "px";
3322
3323     } else if (window.event) {
3324         dragObj.style.left = (dragLeft + window.event.clientX - dragX) + "px";
3325         dragObj.style.top = (dragTop + window.event.clientY - dragY) + "px";
3326     }
3327 }
3328 });
3329
3330
3331 $("body").on("mouseup", function() {
3332
3333     drag = false;
3334 });
3335 document.getElementById("statsdrag").onmousedown = function(e) {
3336     dragObj = document.getElementById("stats");
3337     dragObj.style.position = "relative";
3338
3339     dragTop = parseInt(dragObj.style.top, 10) || 0;
3340     dragLeft = parseInt(dragObj.style.left, 10) || 0;
3341
3342     if (window.event) {
3343         dragX = window.event.clientX;
3344         dragY = window.event.clientY;
3345     } else if (e) {
3346         dragX = e.clientX;
3347         dragY = e.clientY;
3348     }
3349
3350     drag = true;
3351 };
3352
3353 document.getElementById("popupdrag").onmousedown = function(e) {
3354     dragObj = document.getElementById("popup");
3355     dragObj.style.position = "relative";
3356
3357     dragTop = parseInt(dragObj.style.top, 10) || 0;
3358     dragLeft = parseInt(dragObj.style.left, 10) || 0;
3359
3360     if (window.event) {
3361         dragX = window.event.clientX;
3362         dragY = window.event.clientY;
3363     } else if (e) {
3364         dragX = e.clientX;
3365         dragY = e.clientY;
3366     }
3367
3368     drag = true;
3369 };
3370
3371 $("#mortgagebutton").click(function() {
3372     var checkedProperty = getCheckedProperty();
```

```

3373     var s = square[checkedProperty];
3374
3375     if (s.mortgage) {
3376         if (player[s.owner].money < Math.round(s.price * 0.55)) {
3377             popup("<p>You need ATP" + (Math.round(s.price * 0.55) - player[s.owner]
3378 .money) + " more to unmortgage " + s.name + "</p>");
3379
3380         } else {
3381             popup("<p>" + player[s.owner].name + ", are you sure you want to
3382 unmortgage " + s.name + " for ATP" + Math.round(s.price * 0.55) + "?</p>", function() {
3383                 unmortgage(checkedProperty);
3384             }, "Yes/No");
3385         }
3386     } else {
3387         popup("<p>" + player[s.owner].name + ", are you sure you want to mortgage " +
3388 s.name + " for ATP" + Math.round(s.price * 0.5) + "?</p>", function() {
3389             mortgage(checkedProperty);
3390         }, "Yes/No");
3391     }
3392
3393     $("#buyhousebutton").on("click", function() {
3394         var checkedProperty = getCheckedProperty();
3395         var s = square[checkedProperty];
3396         var p = player[s.owner];
3397         var houseSum = 0;
3398         var hotelSum = 0;
3399
3400         if (p.money < s.houseprice) {
3401             if (s.house === 4) {
3402                 popup("<p>You need ATP" + (s.houseprice - player[s.owner].money) + " more
3403 to buy a hotel for " + s.name + "</p>");
3404                 return;
3405             } else {
3406                 popup("<p>You need ATP" + (s.houseprice - player[s.owner].money) + " more
3407 to buy a house for " + s.name + "</p>");
3408                 return;
3409             }
3410         }
3411
3412         for (var i = 0; i < 34; i++) {
3413             if (square[i].hotel === 1) {
3414                 hotelSum++;
3415             } else {
3416                 houseSum += square[i].house;
3417             }
3418         }
3419
3420         if (s.house < 4 && houseSum >= 32) {
3421             popup("<p>All 32 houses are owned. You must wait until one becomes available.
3422 </p>");
3423             return;
3424         } else if (s.house === 4 && hotelSum >= 12) {
3425             popup("<p>All 12 hotels are owned. You must wait until one becomes available.
3426 </p>");
3427             return;
3428         }
3429
3430         buyHouse(checkedProperty);
3431     });

```



```
3428
3429     $("#sellhousebutton").click(function() { sellHouse(getCheckedProperty()); });
3430
3431     $("#viewstats").on("click", showStats);
3432     $("#statsclose, #statsbackground").on("click", function() {
3433         $("#statswrap").hide();
3434         $("#statsbackground").fadeOut(400);
3435     });
3436     document.getElementById("viewstats").addEventListener("click", function() {
3437         console.log("View Stats button clicked!"); // Debugging message
3438         alert("View Stats button clicked!"); // Extra confirmation
3439         showStats();
3440     });
3441
3442     document.getElementById("viewstats").addEventListener("click", showStats);
3443     document.getElementById("statsclose").addEventListener("click", hideStats);
3444
3445     function hideStats() {
3446         console.log("▼ Closing stats popup.");
3447         $("#statsbackground").fadeOut(400, function() {
3448             $("#statswrap").hide();
3449         });
3450     }
3451
3452
3453     document.getElementById("statsclose").addEventListener("click", function() {
3454         document.getElementById("statswrap").style.display = "none";
3455         document.getElementById("statsbackground").style.display = "none";
3456     });
3457
3458
3459
3460     $("#buy-menu-item").click(function() {
3461         $("#buy").show();
3462         $("#manage").hide();
3463
3464         // Scroll alerts to bottom.
3465         $("#alert").scrollTop($("#alert").prop("scrollHeight"));
3466     });
3467
3468
3469     $("#manage-menu-item").click(function() {
3470         $("#manage").show();
3471         $("#buy").hide();
3472     });
3473
3474
3475     $("#trade-menu-item").click(game.trade);
3476
3477
3478 };
3479
3480 document.addEventListener("DOMContentLoaded", function () {
3481     document.querySelectorAll(".cell").forEach(cell => {
3482         cell.addEventListener("mouseover", function () {
3483             let propertyIndex = this.id.replace("cell", ""); // Extract index
3484             showdeed(propertyIndex);
3485         });
3486         cell.addEventListener("mouseout", function () {
3487             hidedeed();
```

```
3488         });  
3489     });  
3490 });  
3491
```